Detector overview, what changes with Phase-2

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INFN Pisa

Why I am here

- Integration and test of Tracker Inner Barrel (2006-2007)
- Tracker commissioning at TIF and P5 (2007-2009)
- SiStrip Tracker local reco WG convener
- Tracking POG convener (2010-2011)
- Tracker DPG convener (2012)
- Tracker Deputy Project Manager and Project Manager (2013-2019)
- Tracker Upgrade Coordinator (2021-2026)

Integration and Commissioning

Run 1

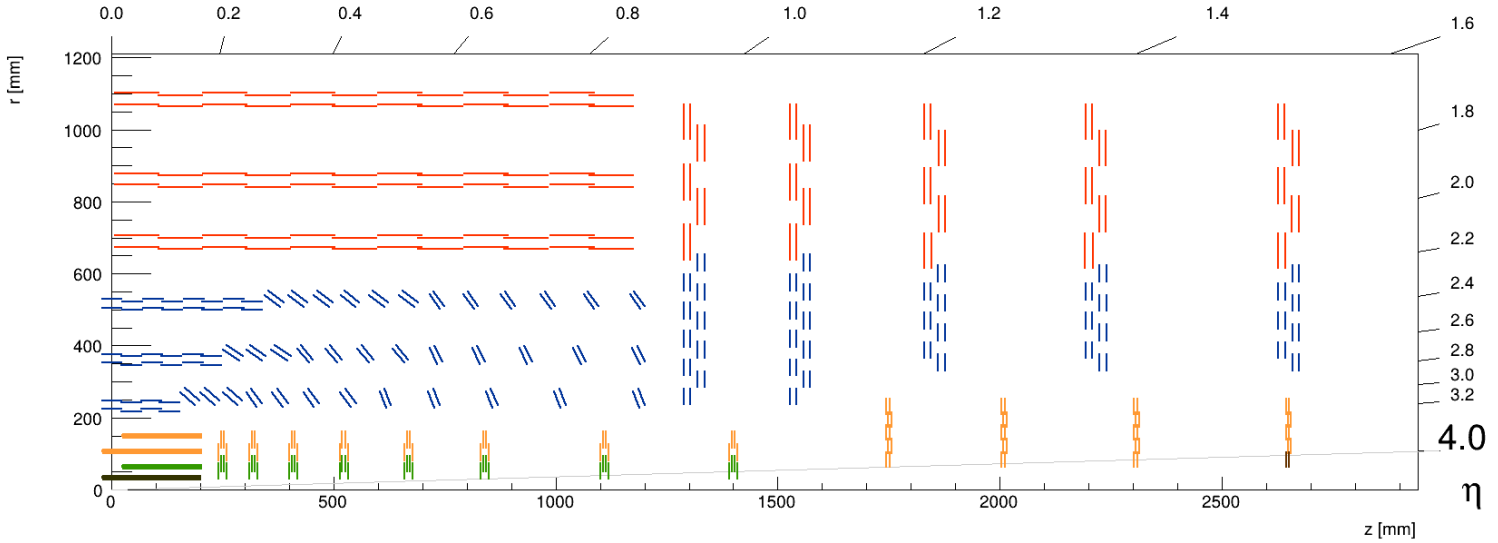
Run 2
Pixel Phase-1 installation

Outline

- List of what is “new” in the Phase-2 Tracker detector and environment
- What (I think it) is important for the DPG because of those new features
- Several “excursions” from the main path for suggestive historical analogies
  - Mostly Run 1 and 2
  - Mostly biased (by me)
  - Mostly NOT with a direct link to possible phase-2 challenges
- Disclaimer
  - Not aiming to cover everything: this workshop agenda is complete and “does not need this presentation”
  - Virtually no time to follow any Phase-2 DPG discussion or activity in the past years (so far!): I may address obvious topics or point to something already ruled out!

Tracker Geometry and Layout

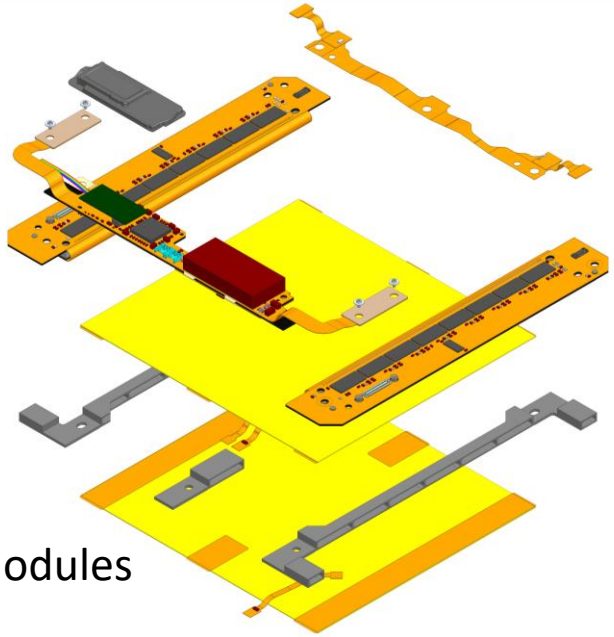
- Two main subdetectors:
  - Outer Tracker: strip(ish)-like, same volume of SiStrip Tracker



  - Inner Tracker: pixel detector, same radii of SiPixel Tracker but with significant forward extension

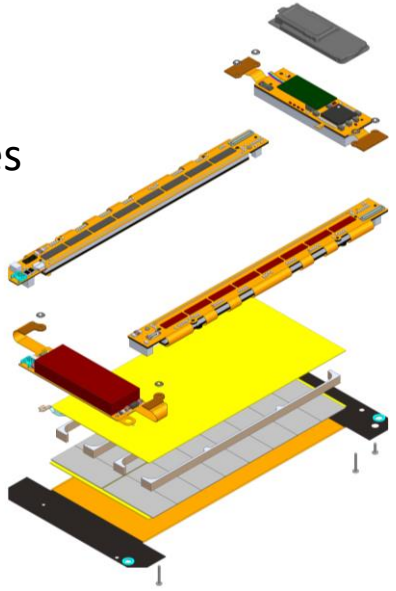
Outer Tracker

- Three subdetectors
  - TB2S: layout and mechanics similar to TOB
    - Modules on “ladders”
  - TBPS: flat and tilted (new concept) regions
    - Modules on “planks” and “rings”
  - TEDD: layers disk-like – petals replaced by “dees” (half disks) as integration units
- Only six layers in the barrel and five disks in endcaps
  - But all modules with two close sensors: two hits each module
- Higher granularity thanks to:
  - Shorter strips: 5 cm (2S) and 2.5 cm (PS)
  - Macro pixels: 1.4 mm (PS)



2S modules

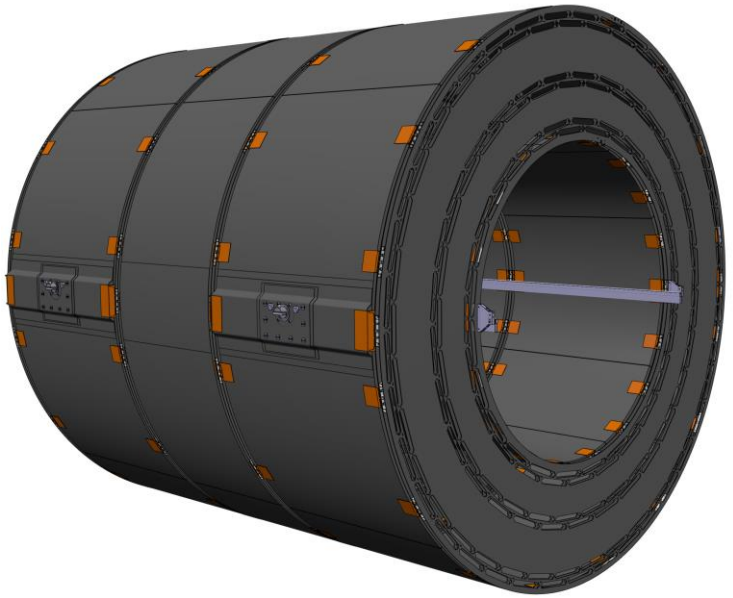
PS modules

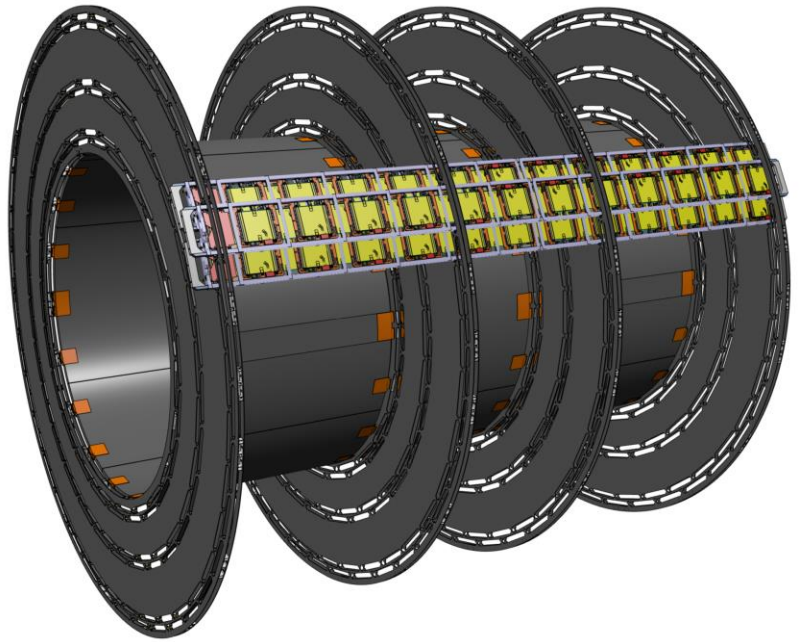


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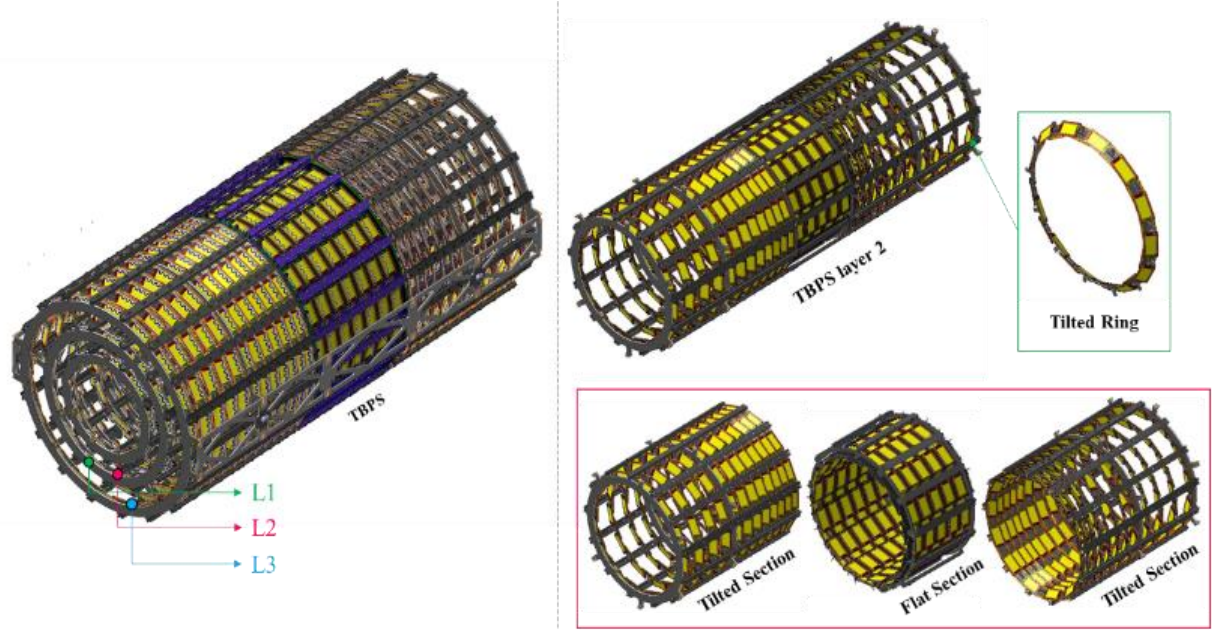


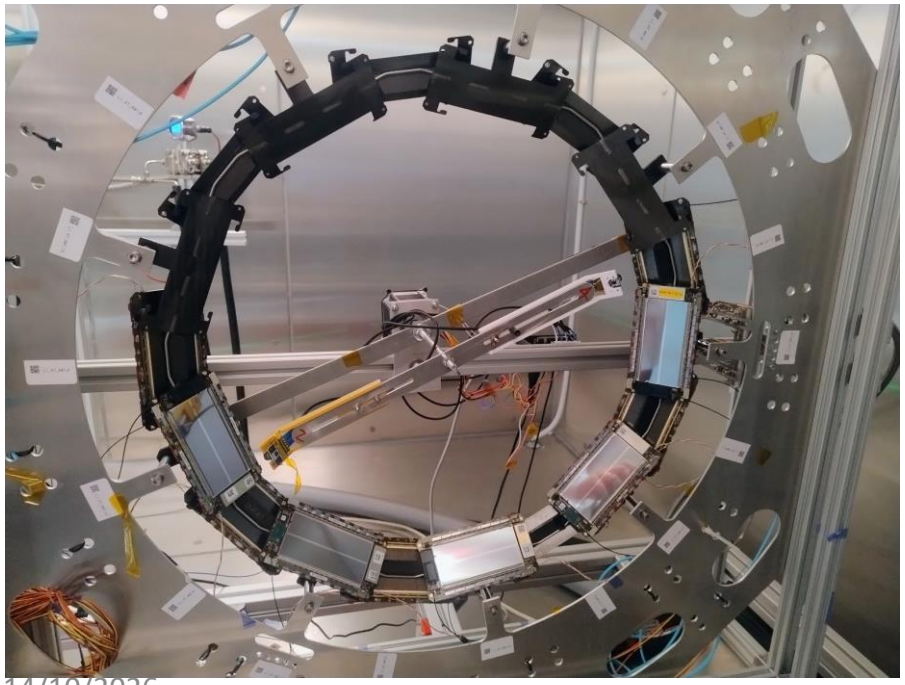




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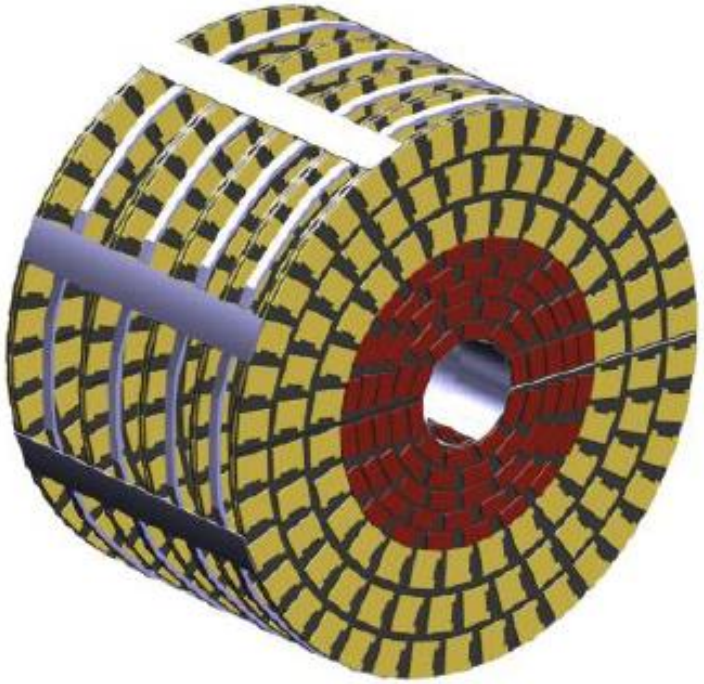


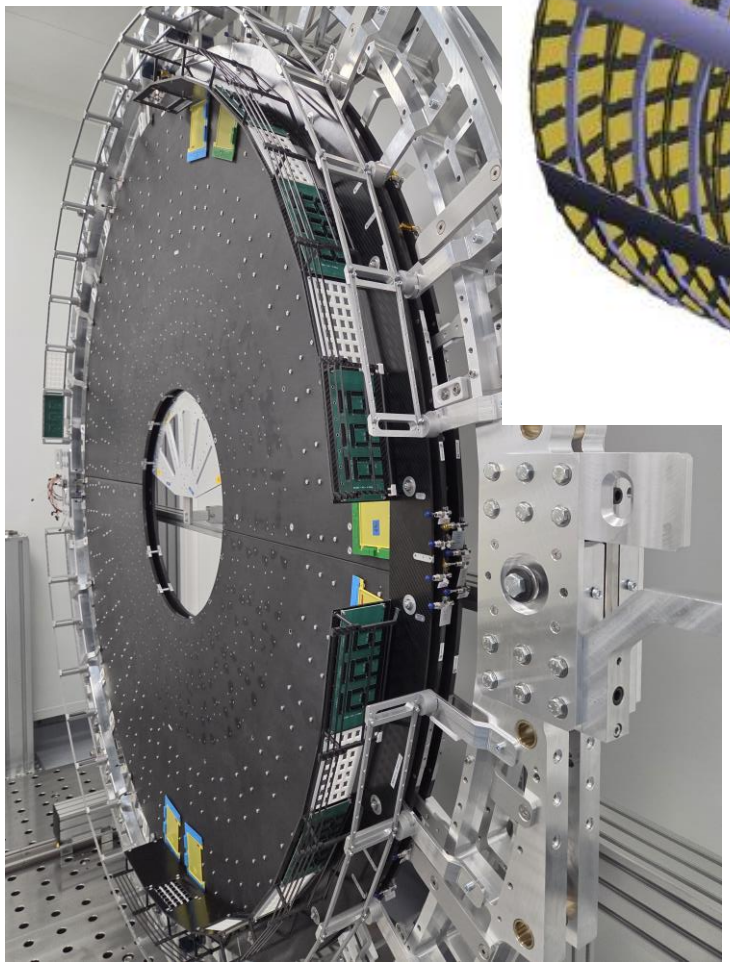


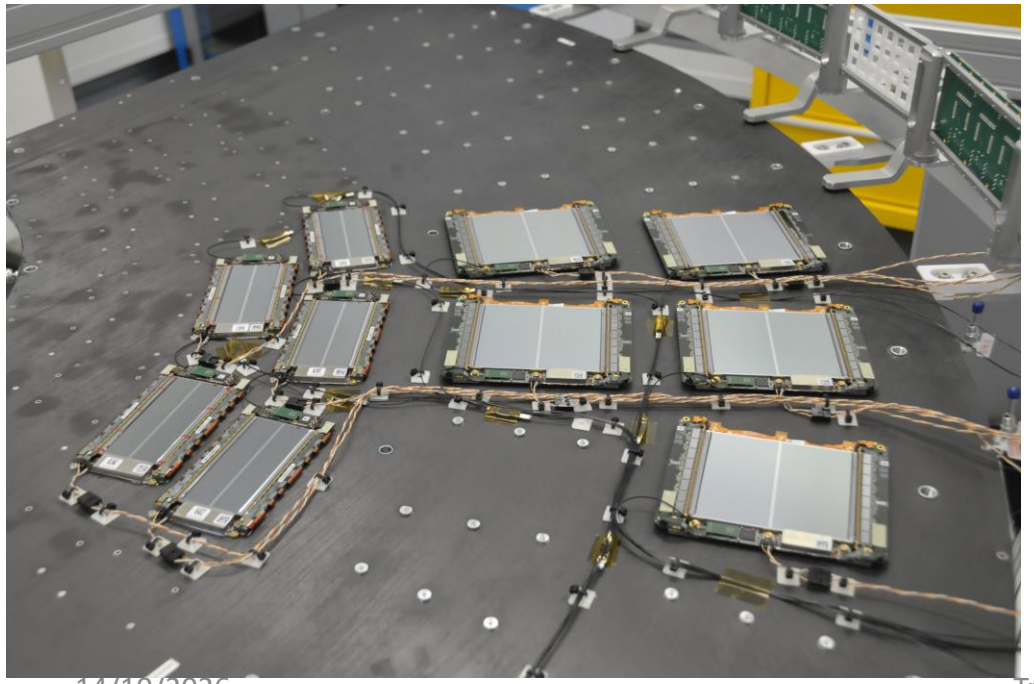


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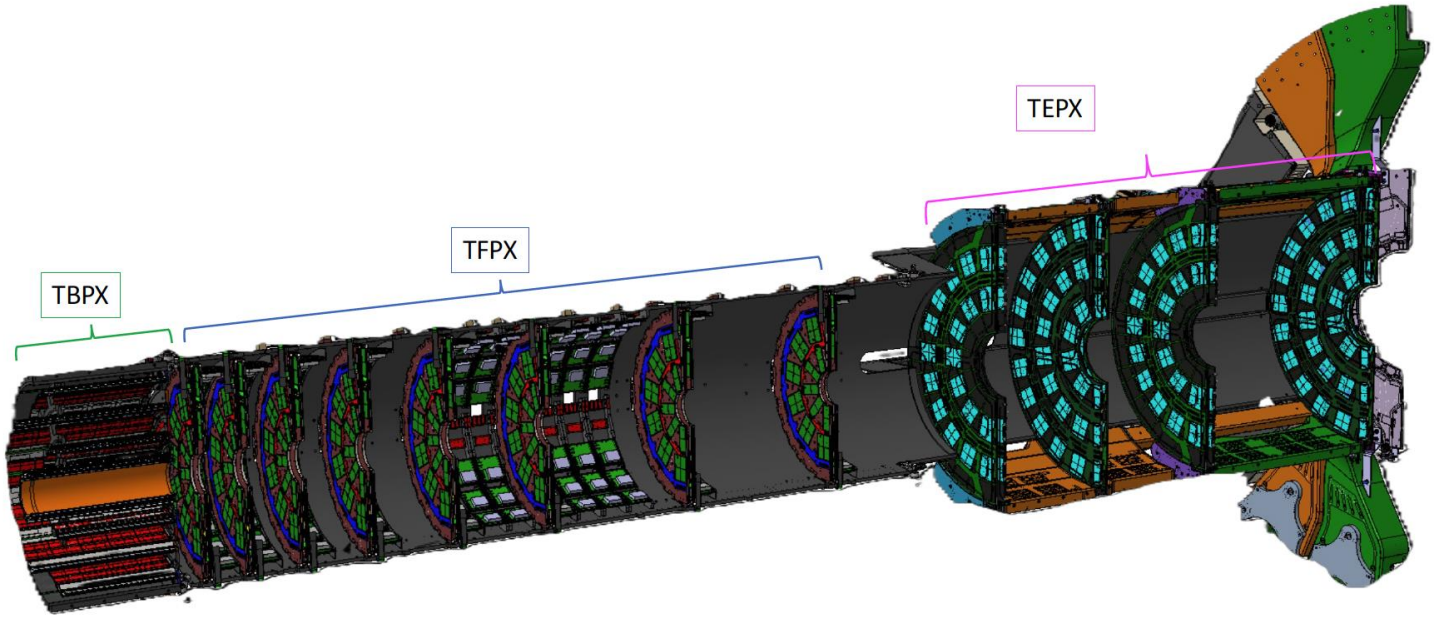






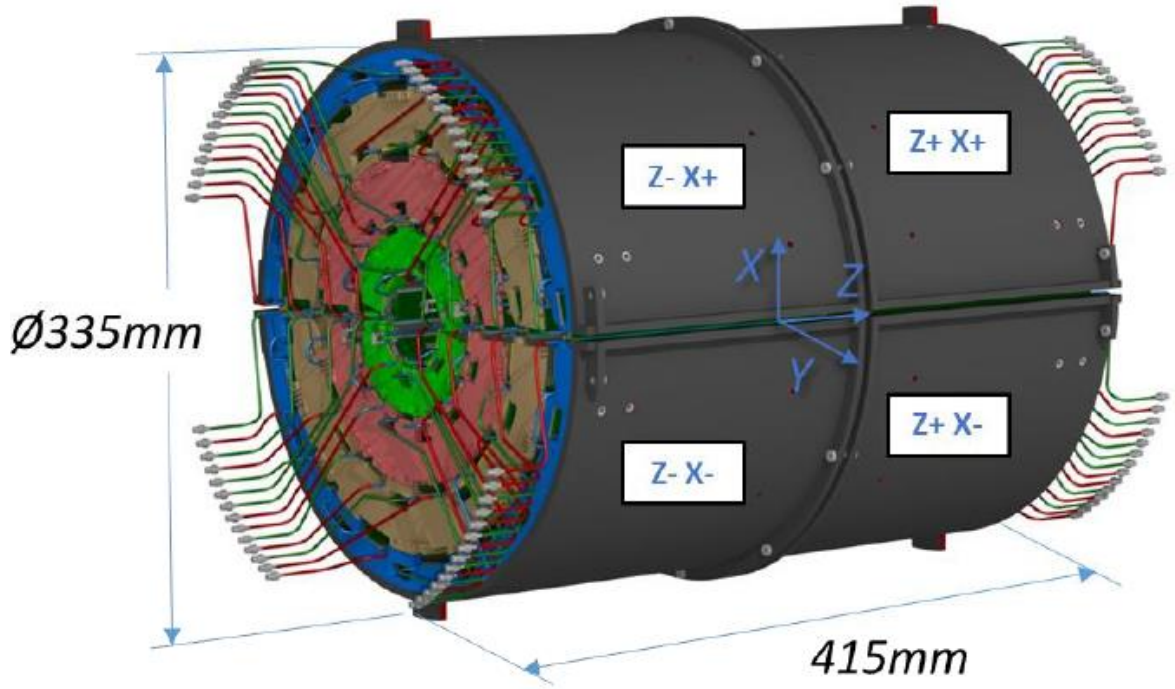
Inner Tracker

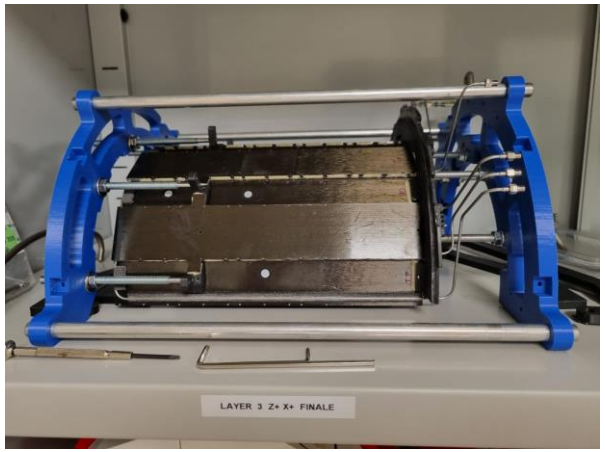
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  - TBPX: volume and mechanics similar to BPIX
    - Four layers, split in four quarters
    - Layer 1 inner radius as in phase-1 BPIX: 27.5 mm
      - Was 44 mm in legacy pixel detector
  - TFPX: aiming to extend forward acceptance
    - Eight disks on each side (z+/-)
    - Same “dee” concept as in TEDD
  - TEPX: complete forward acceptance extension
    - Four disks on each size, same “dee” concept
- Smaller pixels, larger granularity
  - 25x100 um (was 100x150 um)

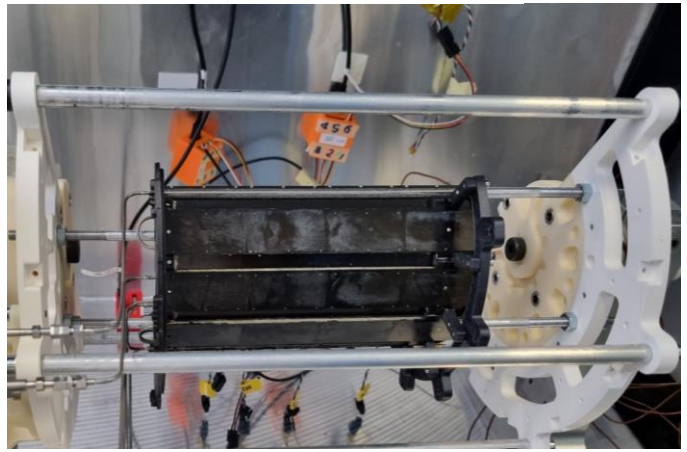


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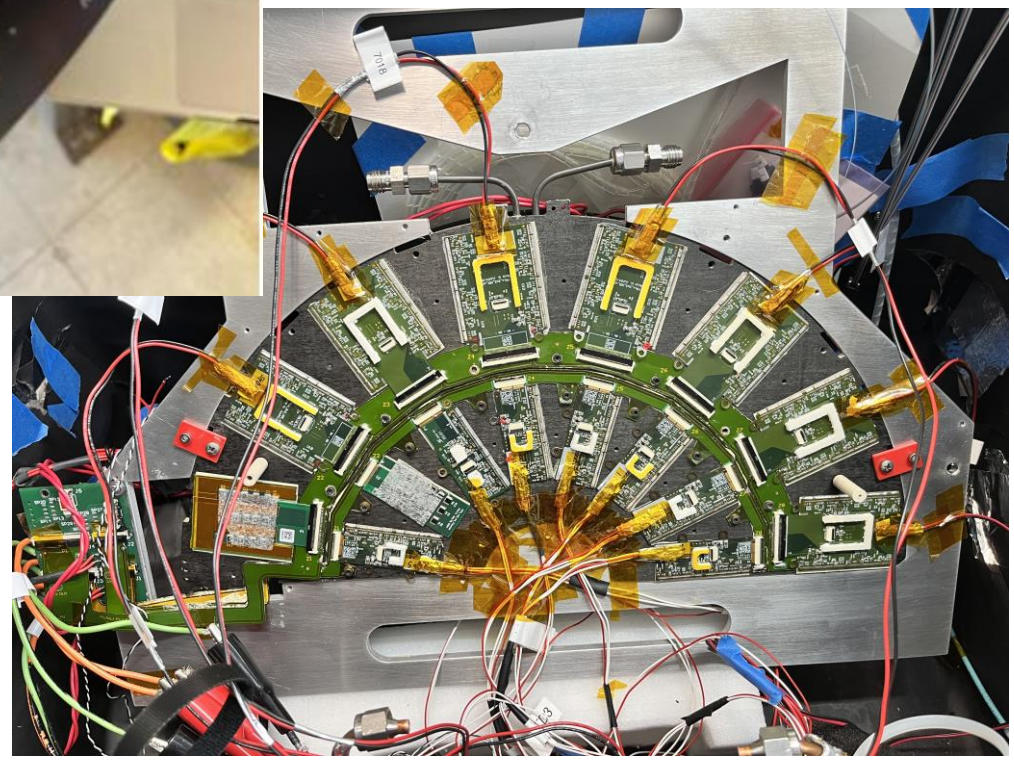




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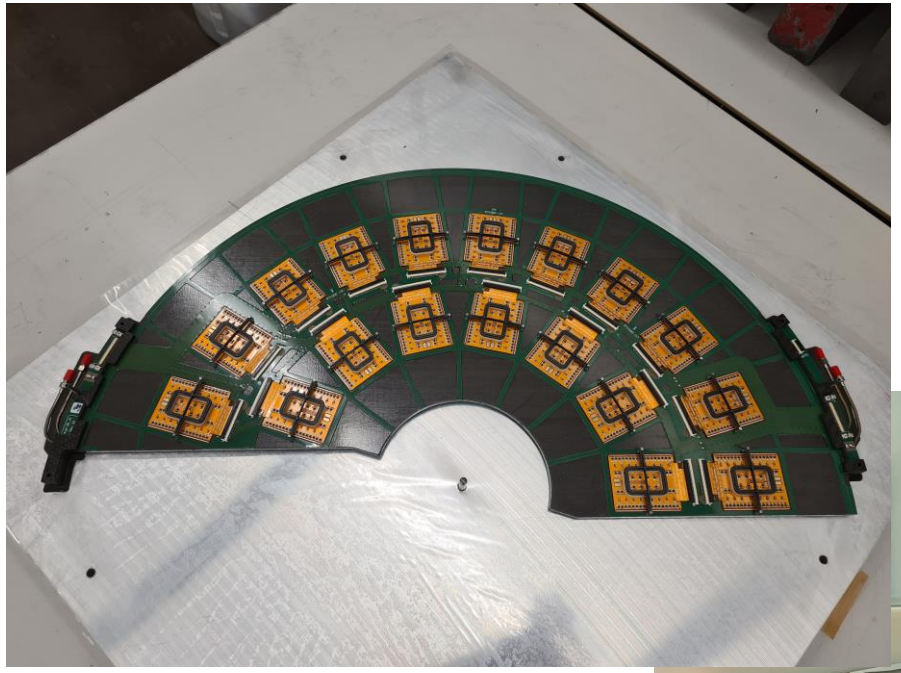
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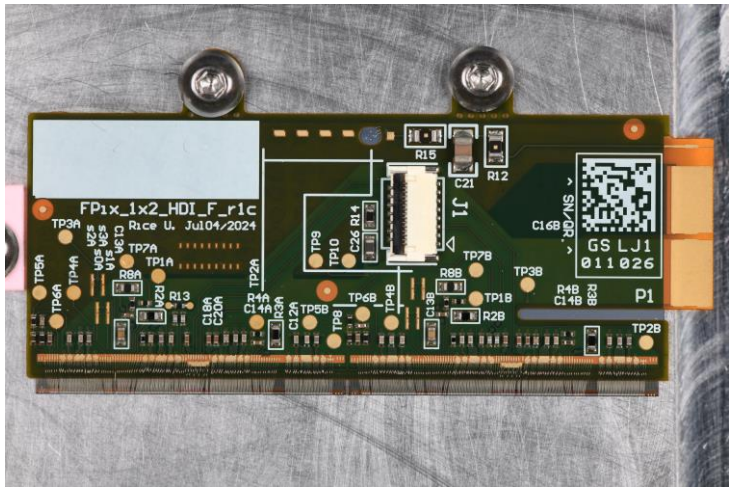


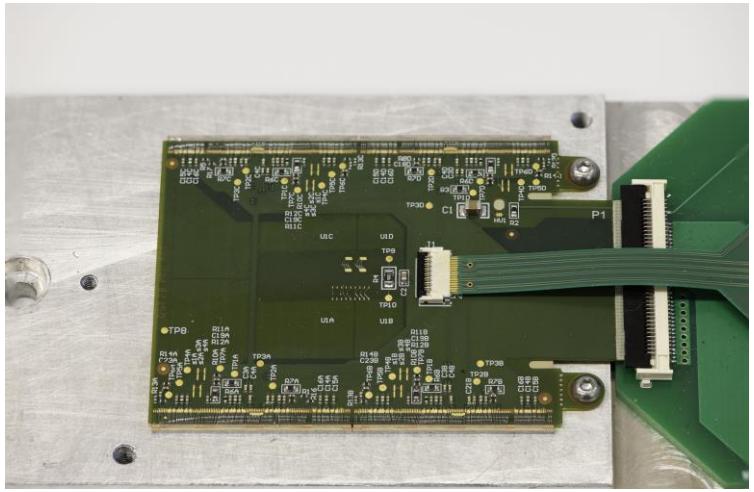


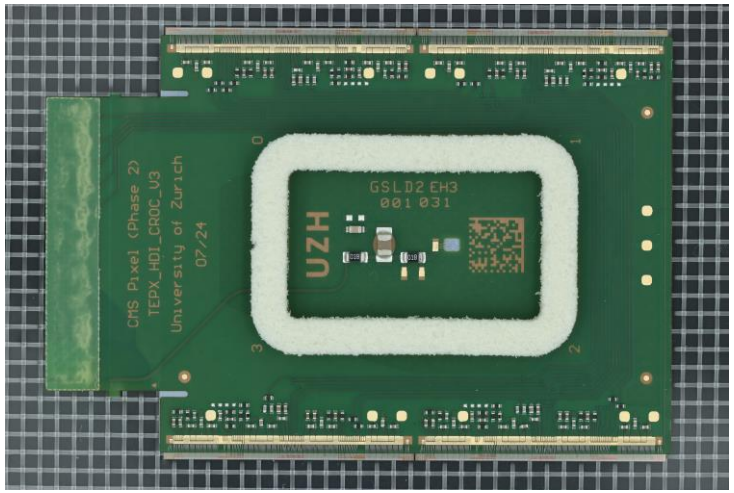
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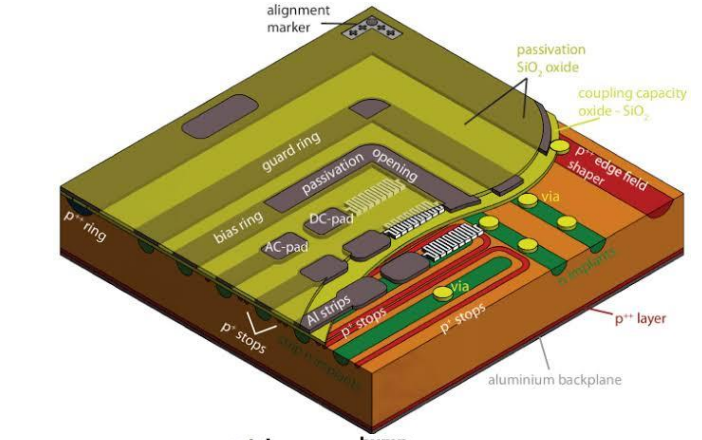


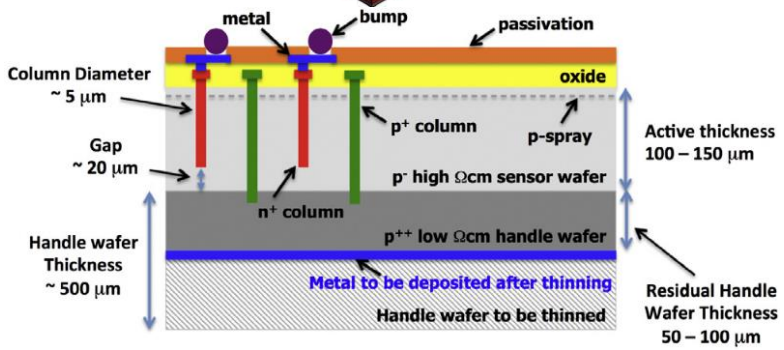


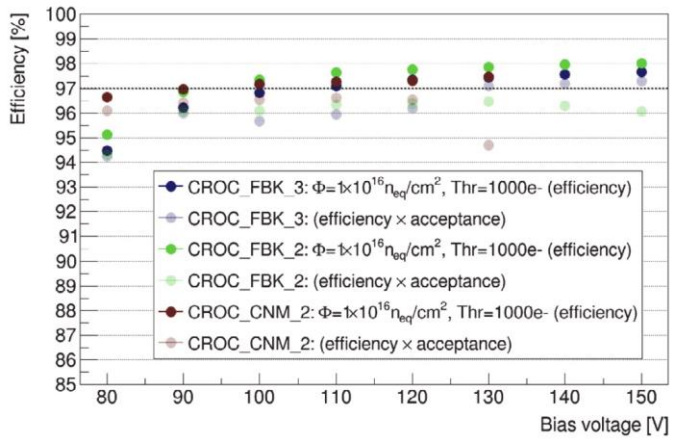


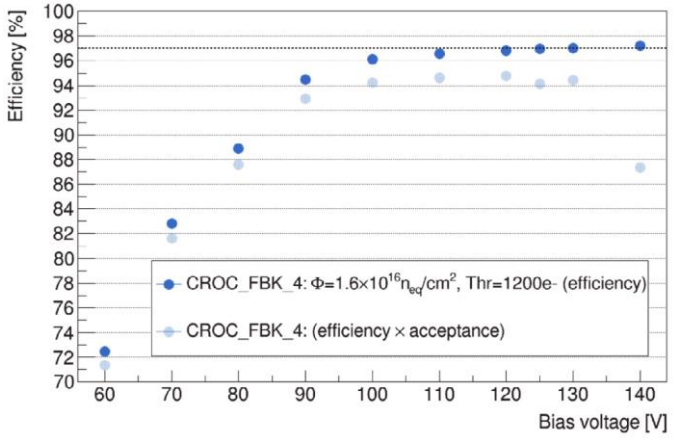
Sensors

- n-in-p material used both in OT and IT
  - Used to be p-in-n in SiStrip Tracker and n-in-n in SiPixel Tracker
  - Segmented signal generated by electrons
    - Electron-like Lorentz Angle
- 3D sensors in TBPX layer 1
  - Lower depletion/bias voltage → easier thermal management
  - Different charge collection pattern
  - Different evolution with radiation
    - Observed to become noisy at about 1e-16 n_{eq}/cm^2



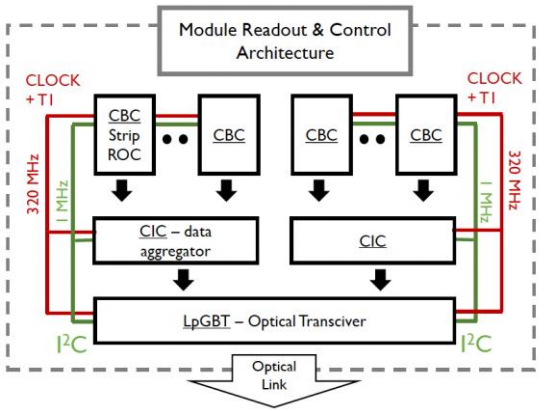


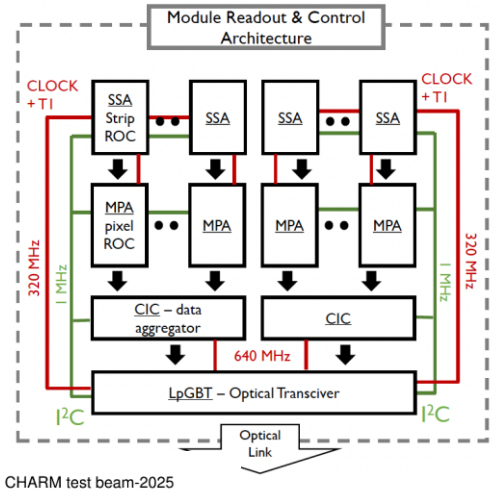


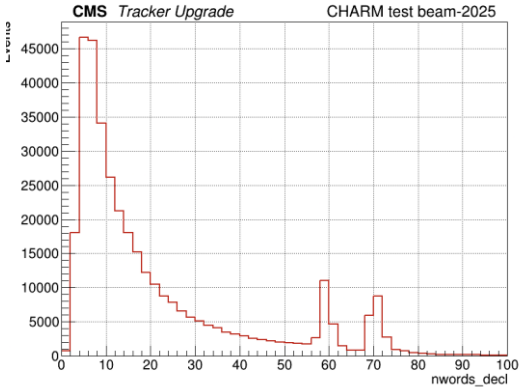


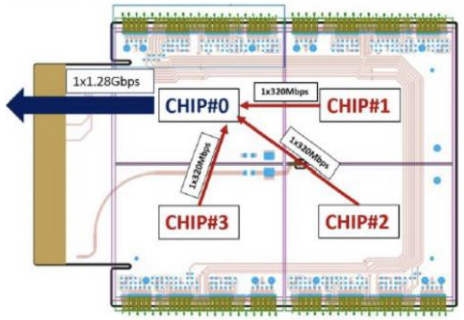
Readout chips

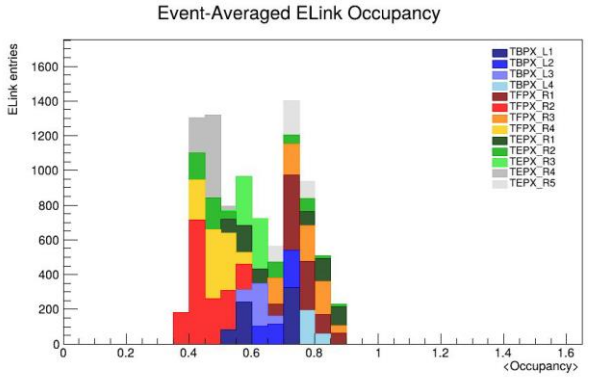
- New readout chips in OT and IT
- OT
  - 2S modules (5Gbps): 16xCBC → 2xCIC → lpGBT → DTC (Serenity)
  - PS modules (5-10 Gbps) : 16xSSA → 16xMPA → 2xCIC →lpGBT → DTC (Serenity)
- IT
  - Quad or Dual modules: 2-4 CROC → lpGBT → DTC (Apollo)
    - Several combinations of data merging across CROC and multi-lines within a CROC
      - Depending on expected data rate
- All chips and readout links designed to be just right at the largest design average luminosity and pileup





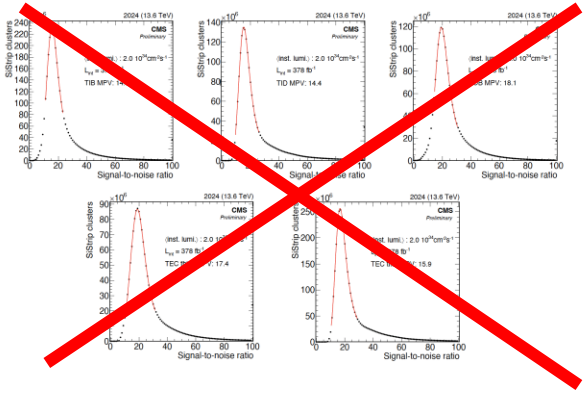


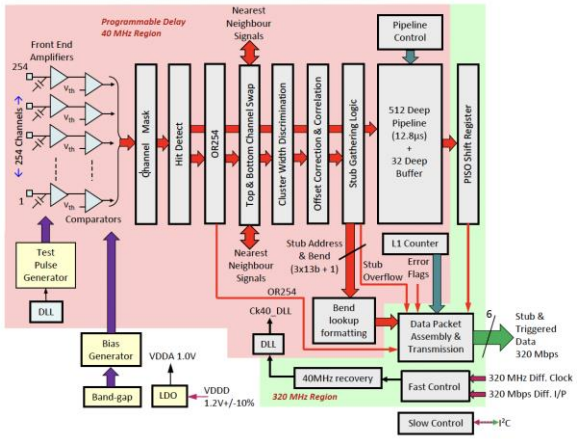


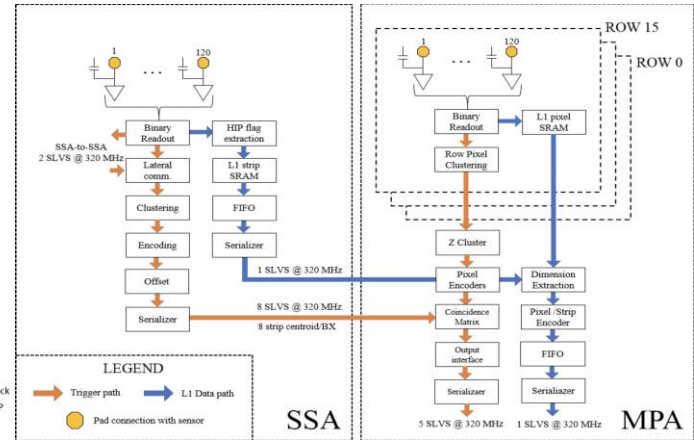


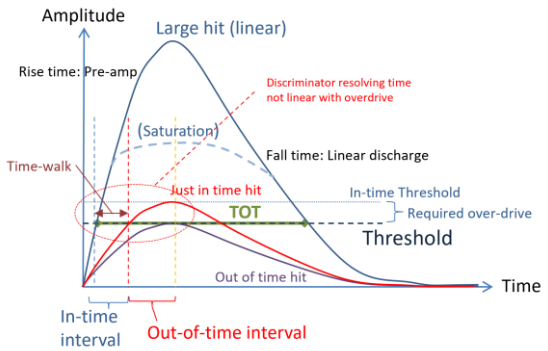
Signal type

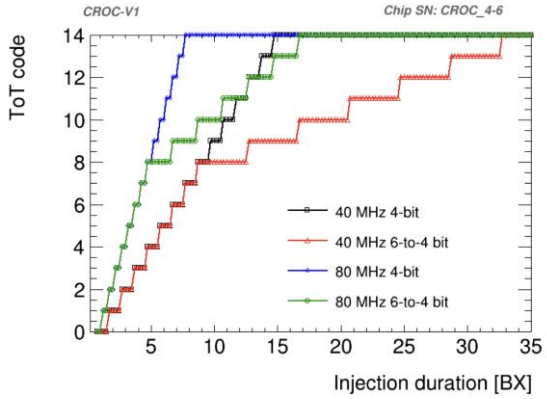
- Outer Tracker
  - Binary signal from all channels of all modules
    - YES/NO above threshold
    - Second threshold to discriminate non-MIP signals in SSA: PS module strip sensors
    - No more G1 and G2!
  - Only front-end-chip thresholds
    - No backend or offline thresholds
- Inner Tracker
  - Time-over-threshold analog-to-digital signal conversion
  - Two different slopes depending on the range





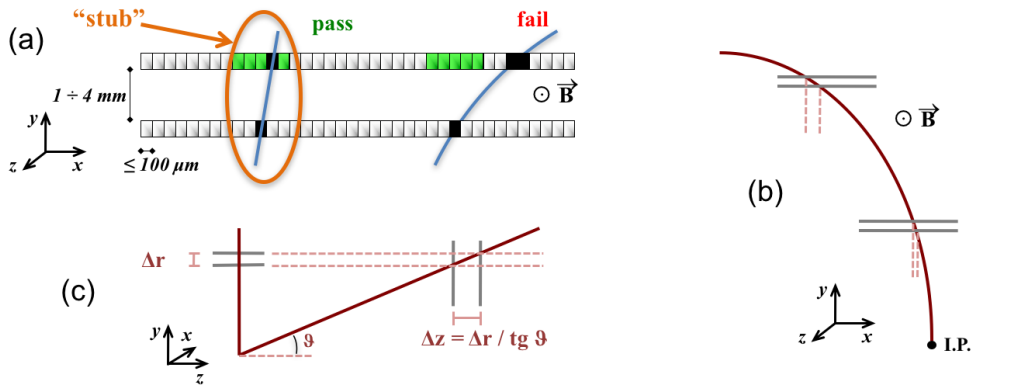


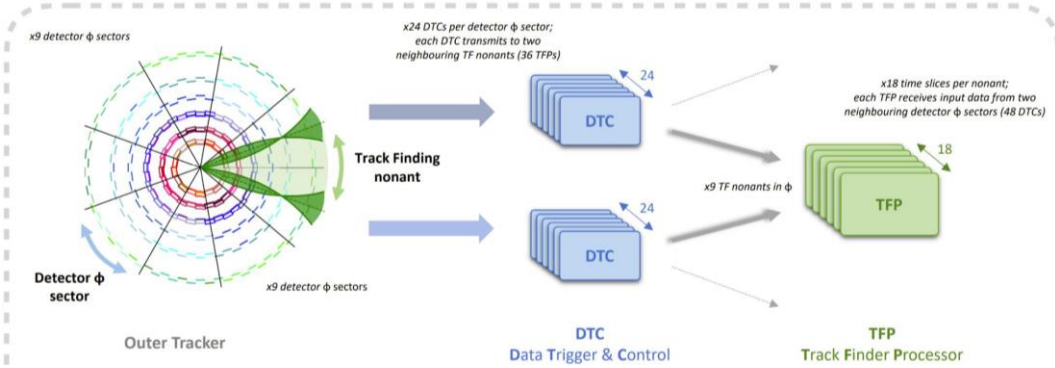


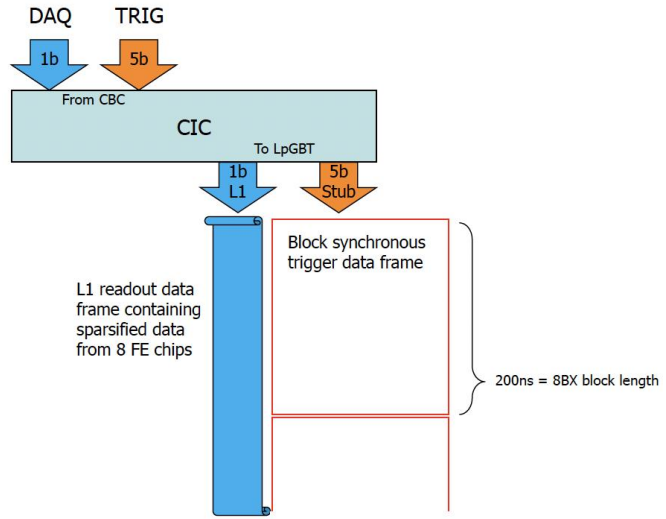


Outer Tracker Stubs

- Subset of hit pairs from OT modules readout every BX
  - Needed to reconstruction tracks as Tracker L1 trigger primitives
  - Completely new perspective with respect to legacy Tracker
- Hit pairs selected by on-module readout chips
  - Taking advantage of sensor separation and magnetic field track bending
  - Several parameters to be configured to define search windows and offsets
- Normal data flow: modules → DTC → Track Finder boards (Apollo)
  - About 80% of data links reserved to stub data
  - There should be “spying” channels/paths to monitor stub data offline

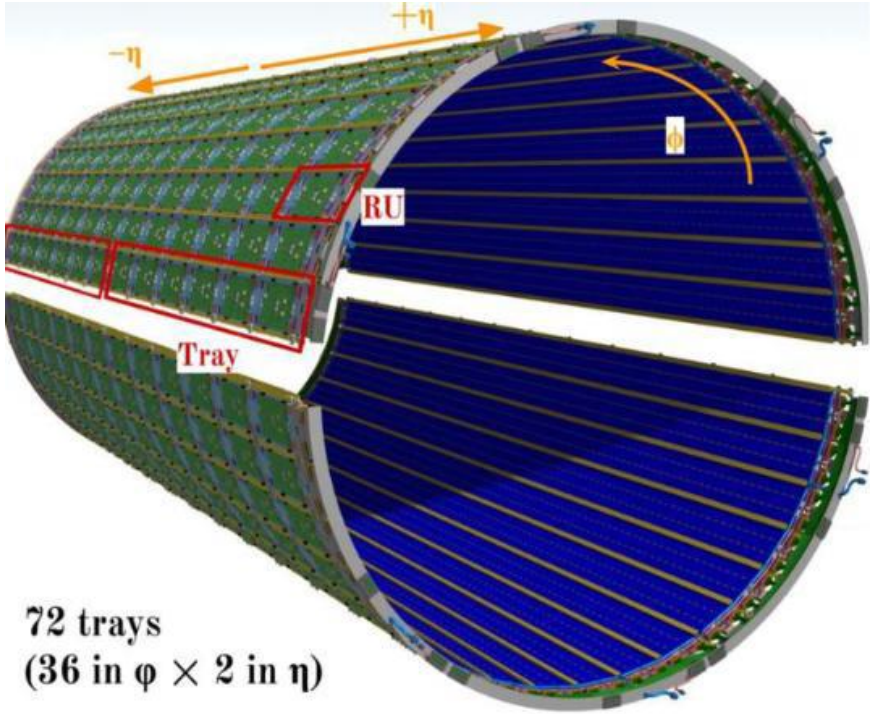






MIP Timing Detector

- New entry among CMS subdetectors
- Barrel Timing Layer hosted inside the BTL-Tracker Support Tube and sharing service channels with the Tracker
- Aiming to assign timing information to charged particles with a resolution of 30-60 ps





Detector and layout brief recap

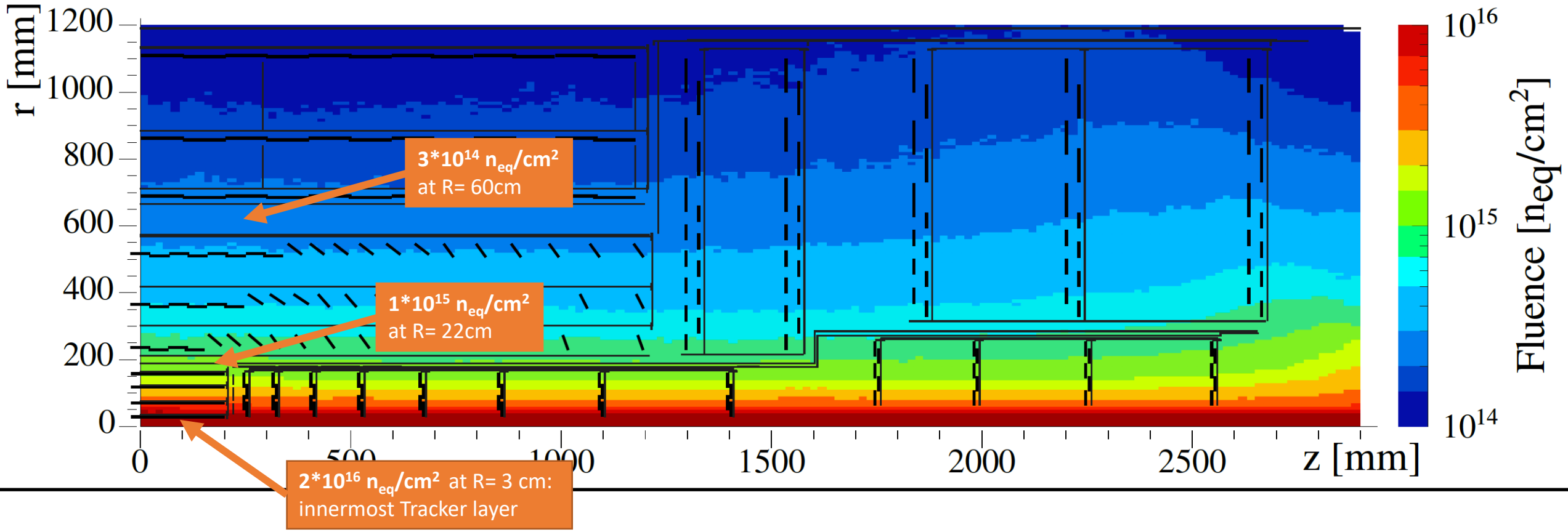
- A few new layout and mechanical structure solutions
- Significant forward extension and large pixel-only acceptance regions
- TBPX layer 1 as close to beam as phase-1 (and closer than legacy pixel detector)
- Different planar sensor technology and 3D sensors in TBPX layer 1
- New readout chip and chain
- Binary signal with on-front-end threshold from Outer Tracker
- Triggerless 40 MHz stub data from Outer Tracker and L1 Trigger Track reconstruction
- MIP Timing Detector

HL-LHC conditions

- Potential peak luminosity: > 1e35 $cm^{-2}s^{-1}$
- Likely physics-run peak luminosity: 5-7.5e34 $cm^{-2}s^{-1}$
  - With luminosity levelling
- Pileup: 140-200
  - Depending on peak luminosity and number of bunches
- Total integrated luminosity: 3000 - 4000 fb^{-1}
- Detector designed for PU 200 and 4000 fb^{-1}
- About 4 (instantaneous) and 10 (integrated) times larger than so far

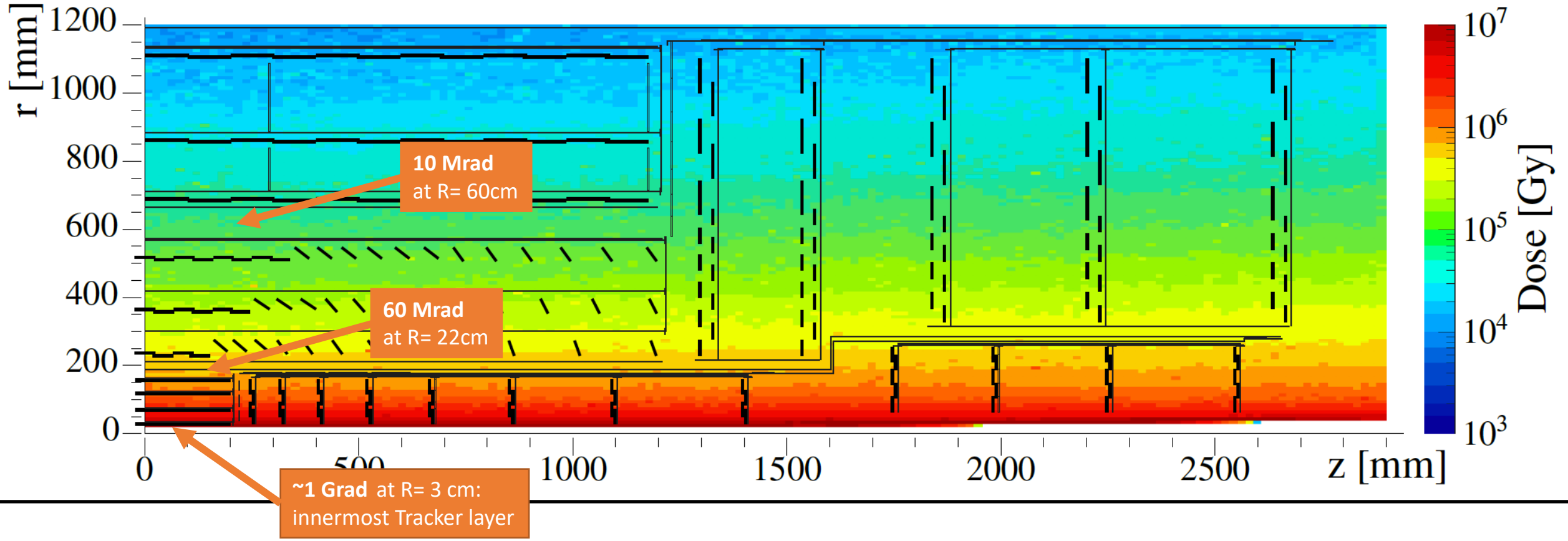
Disclaimer:
8-year old slide of mine!

1 MeV n_{eq} fluence after 3000 fb^{-1}



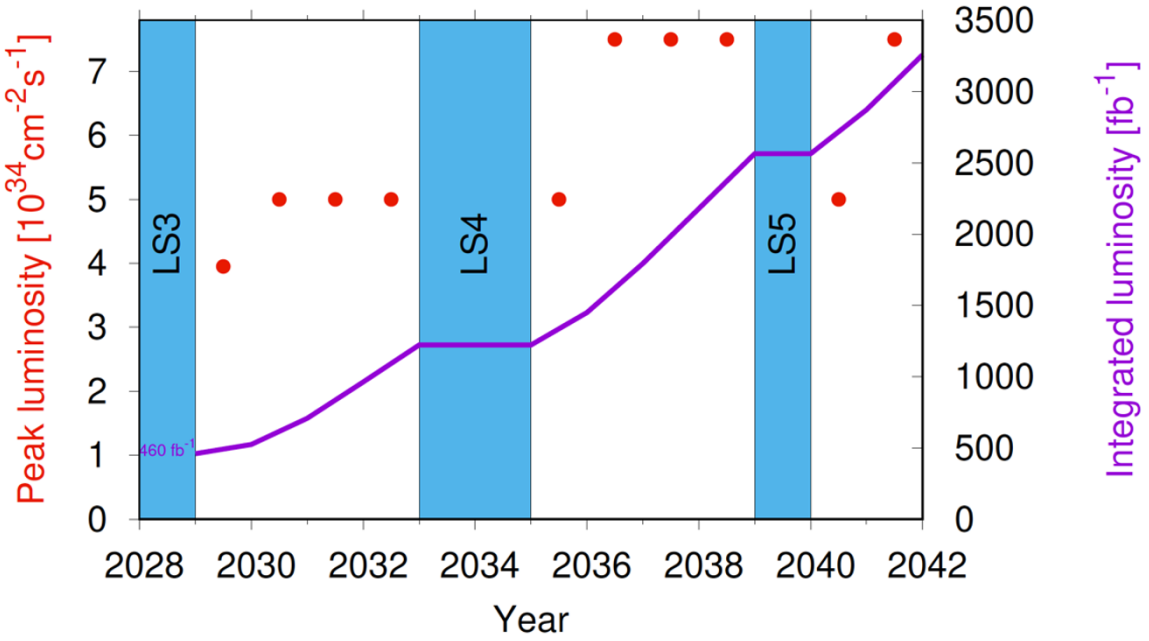
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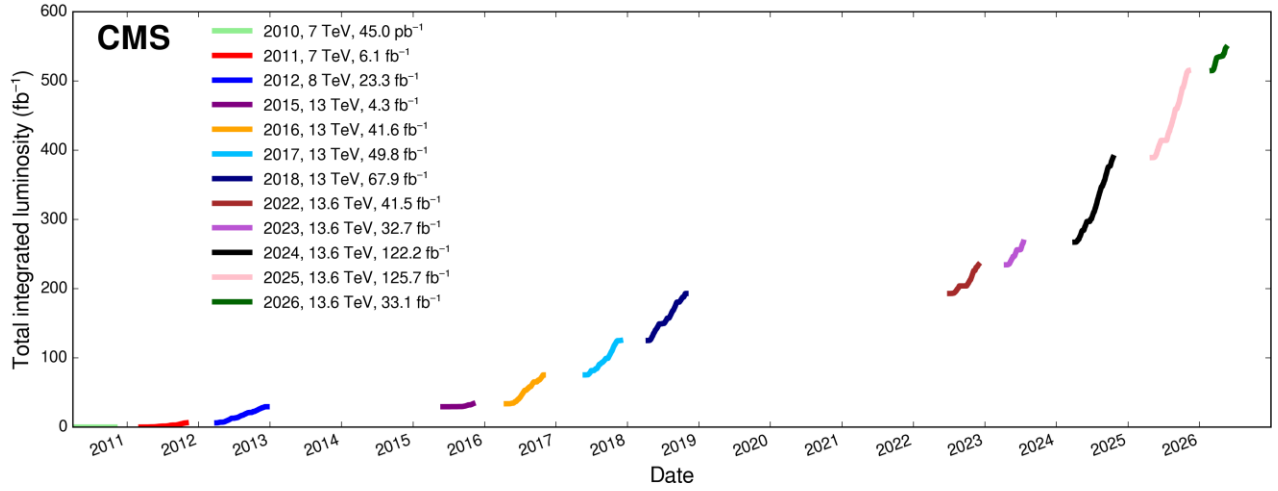
Integrated dose after 3000 fb^{-1}

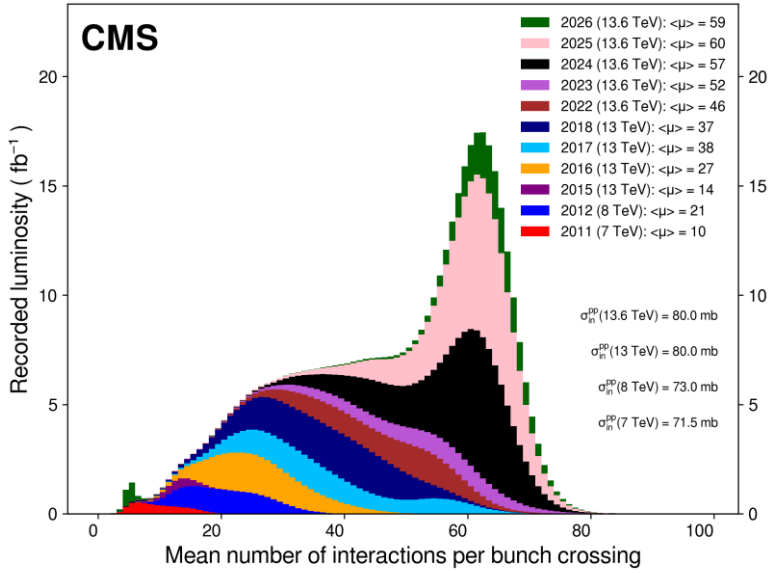


HL-LHC time profile

- 4e34 $cm^{-2}s^{-1}$ during first year
- 5e34 $cm^{-2}s^{-1}$ during Run 4
- 7.5e34 $cm^{-2}s^{-1}$ in Run 5
- Nominal ramp up more accelerated than Run 1+2+3

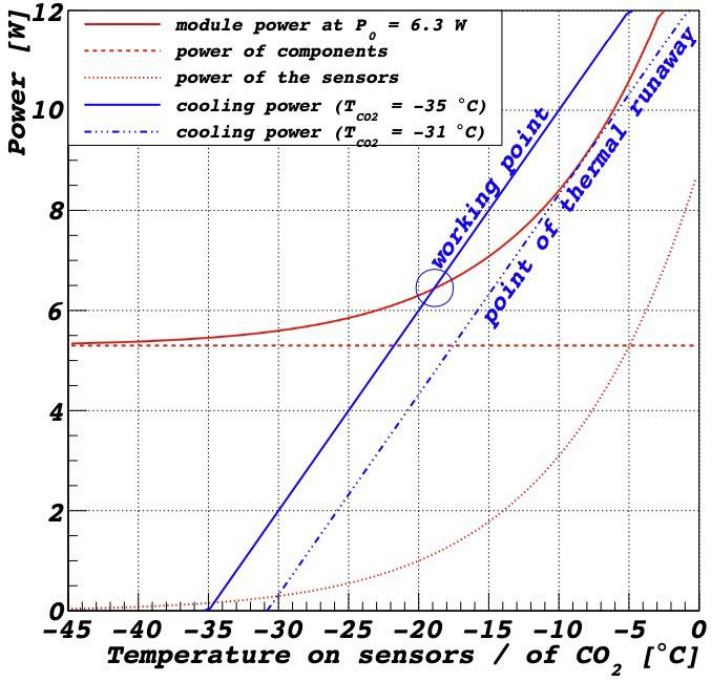


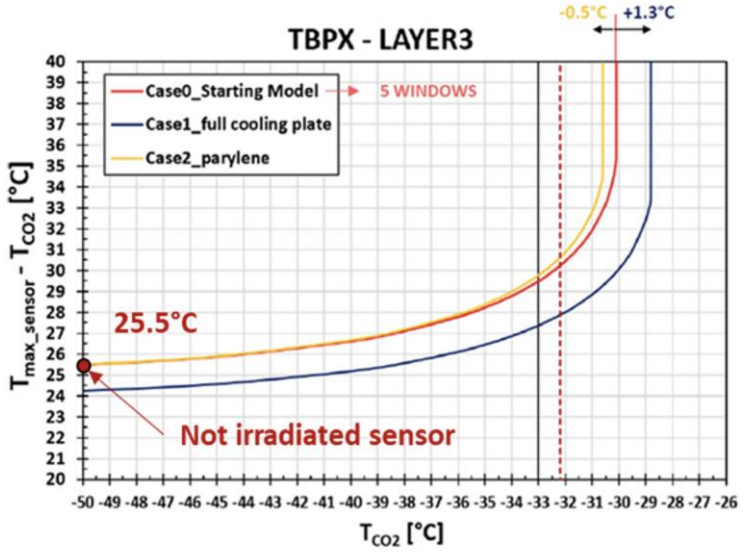




Cooling

- Due to larger irradiation and faster luminosity ramp up Tracker will be operated cold (-35C) since day 1
  - Legacy Tracker was operated at -20C only starting from Run2
- 2-phase CO_2 cooling adopted both for Outer and Inner Tracker
  - So far used only for phase-1 pixel detector: about 10 times more power
- Modules, mechanical structures, and cooling pipe system designed to be just right for the expected luminosity and radiation
  - More exposed and power-dissipating regions with sensible but not huge margin against thermal runaway.





Commissioning

- Outer Tracker schedule is very tight. Presently no time to operate the full detector at TIF after TEDD installation
  - 5 weeks between end of TB2S integration and start of TBPS insertion
  - 11 weeks between one TEDD and the other TEDD installation
- Present Technical Coordination schedule foresees 2+2 weeks for CRUZET+CRAFT
  - 10 months after Outer Tracker installation and 5 months after Inner Tracker installation
    - Time available for local running and trigger availability not known by me
- To be compared to legacy detector
  - 4-5 months at TIF (SiStrip Tracker only): three papers
  - > 1 year at P5 – due to LHC accident: three papers

Running conditions and operation brief recap

- Much larger luminosity, pileup, radiation
- Planned faster ramp-up
  - No 50ns bunch spacing fills, I guess
- New cooling system and no warm operations
- Planned much less time for commissioning and alignment once CMS is ready and before beam

Hit efficiency and occupancy

- Critical tools for detector tuning and monitoring
- “only” tool for Outer Tracker because of binary readout
  - Efficiency instead of signal amplitude to monitor signal collection
- Threshold setting will be critical because on front-end electronics
  - No way to recover offline
  - Trading among efficiency and noise
    - Optimal working point to be studied and defined
      - With MC? Realistic noise model in simulation could become useful to be prepared in advance
- Excursion: TOB correlated noise

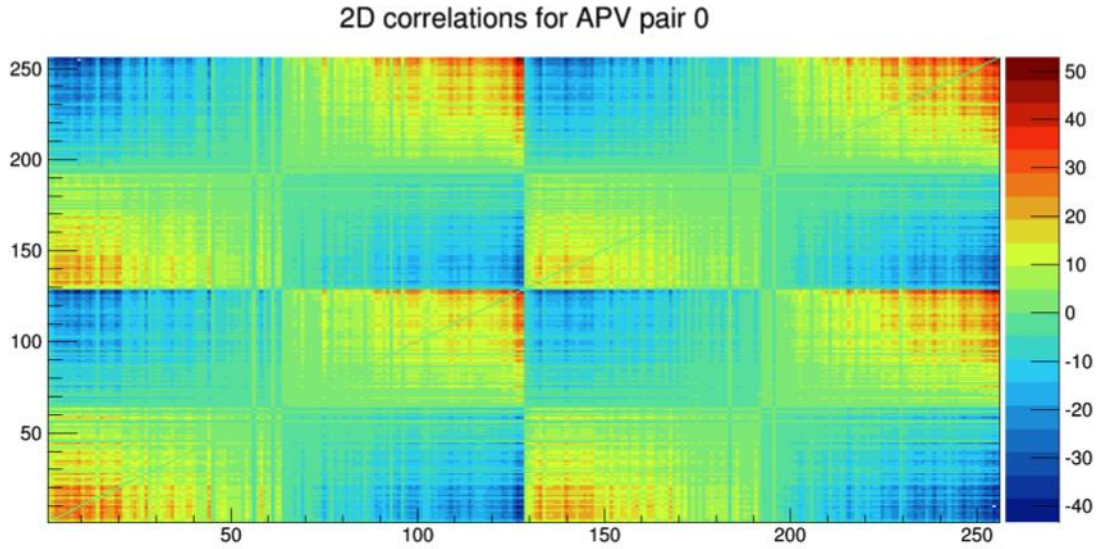
TOB correlated noise: a lesson to be learned

- A sizeable fraction of strips at the edge of APVs have often been masked as noisy in TOB modules at the edge of ladders due to too large occupancy
- Strips affected by correlated noise (aka not-flat common mode)
  - Correlation was known and could be measured quite accurately using Pedestal and Noise run data
- Masking could be avoided if 5σ cluster noise (offline) cut were applied by taking into account correlation:

- Cluster noise: no correlation vs with correlation

$$\sqrt{\sum \sigma_i^2} < \sqrt{\sum \sigma_i^2 + \sum 2\rho\sigma_i\,\sigma_j}$$

- Trading larger threshold against fully masked/inefficient channels

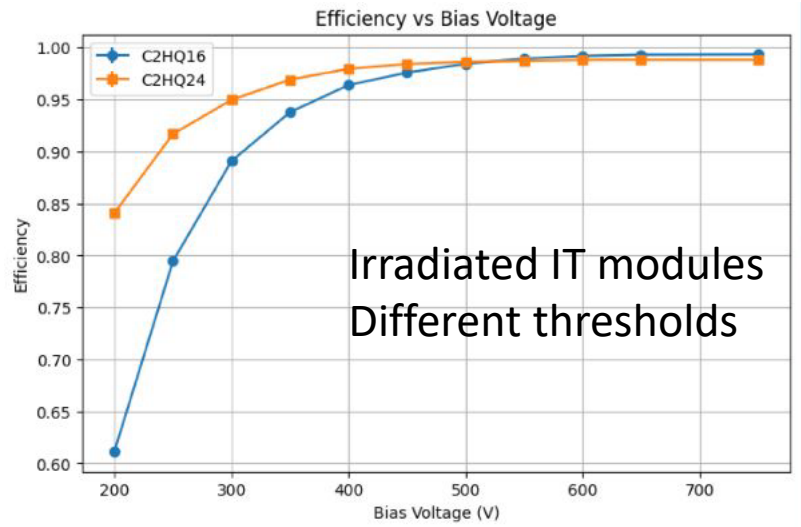


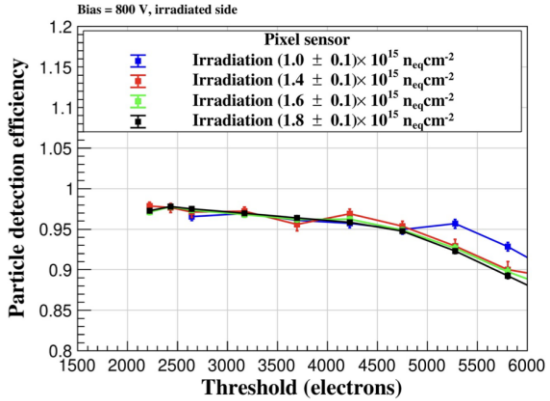
Hit efficiency and occupancy

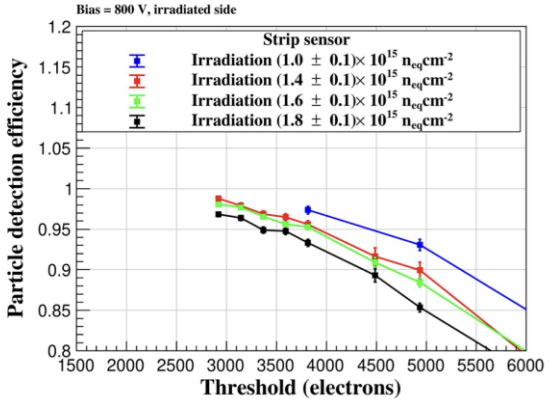
- Important studies enabled by hit efficiency and occupancy measurements (biased/incomplete list)
  - Detector front-end timing and Out-Of-Time pileup optimization and mitigation
  - Characterization of (correlated) noise sources at P5
  - Characterization of ”dynamic” inefficiencies of new chips and readout links
    - Dependencies on trigger rate, luminosity, beam background, trigger sequences, …
    - Simulation (?)
  - Sanity checks of readout chips and links
    - We are already aware of affordable (?) issues in some CBC chips
  - Sanity checks and possible improvements of back-end firmware

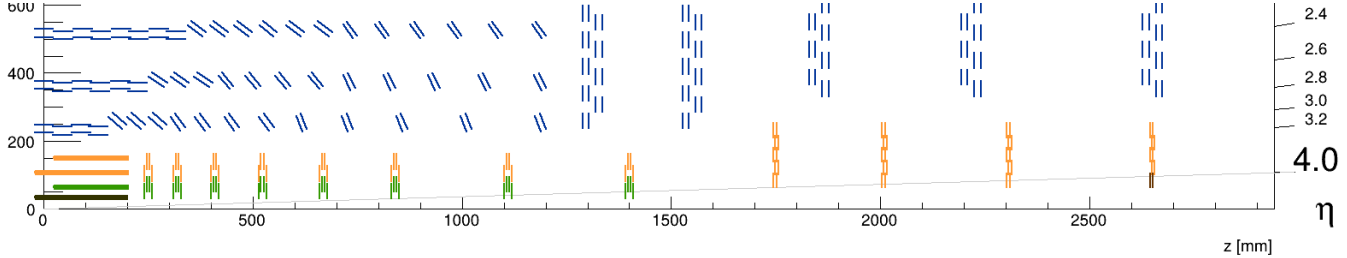
Hit efficiency and occupancy

- Characterization of efficiency and noise vs radiation
  - Losses are expected both in IT and OT sensors
  - 3D sensors are expected to become noise at about 1e16 n_{eq}/cm^2
  - Radiation gradient in TFPX and TEPX modules
- Analysis tools
  - Develop tracking and hit search window strategy for unbiased hit efficiency
  - Pixel-only forward tracking in new acceptance region is a must to exploit TFPX and TEPX
  - Unbiased tracking and hit search in Inner Tracker Layer 1 is a must
    - Usually measuring efficiency of first or last hit on a track is complicated
  - Simulation of radiation damage
    - Let's exploit MC truth completely
  - Optimization of local reconstruction to get best resolution
- Excursion: you will be able "to see everything!"







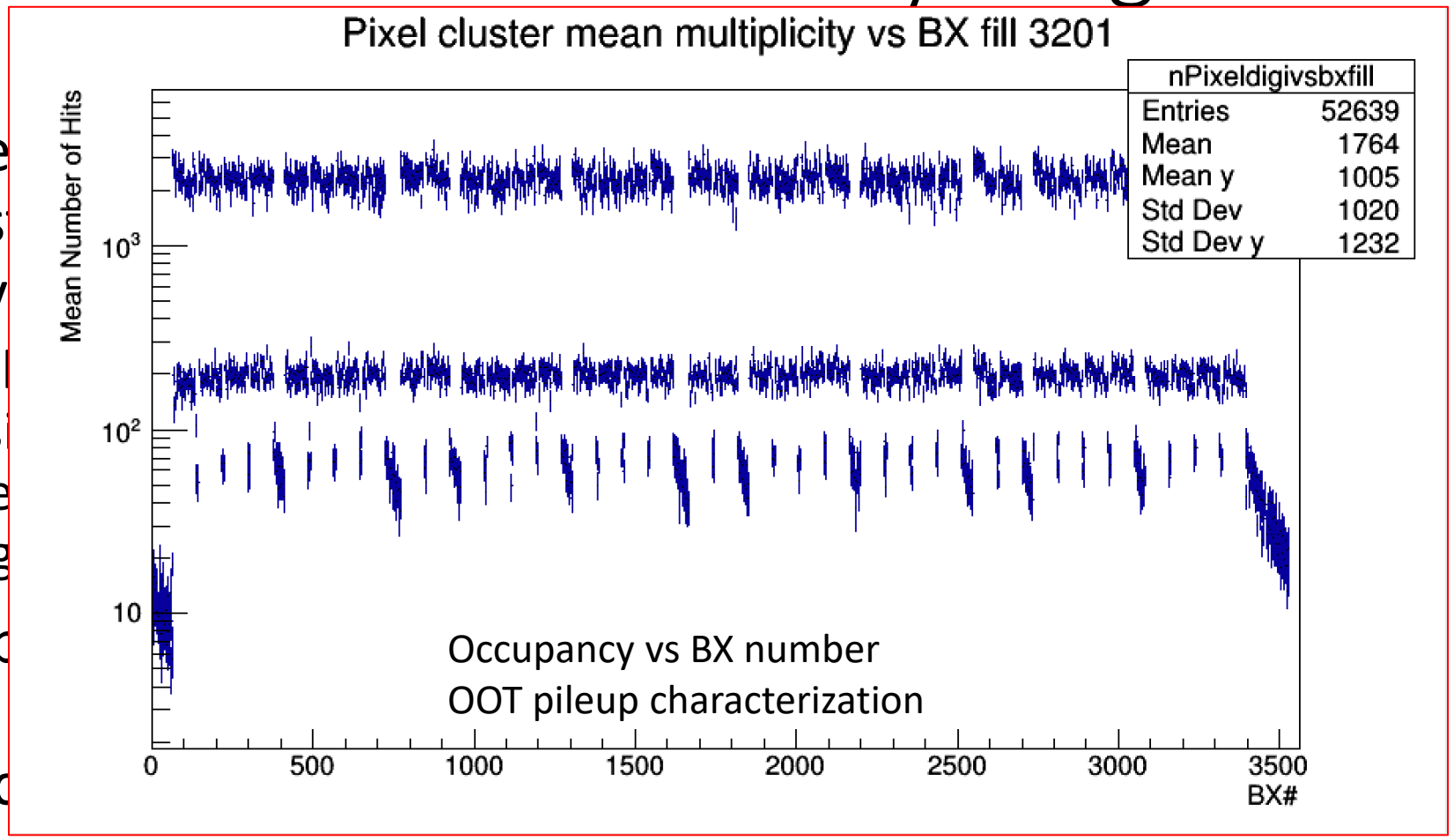


You will be able to see everything

- Number of modules and large event statistics will allow you to see “everything”
  - Much more than what will be possible during prototyping, system tests, production, and integration
- Be prepared to develop tools aiming to exploit this extreme statistical accuracy

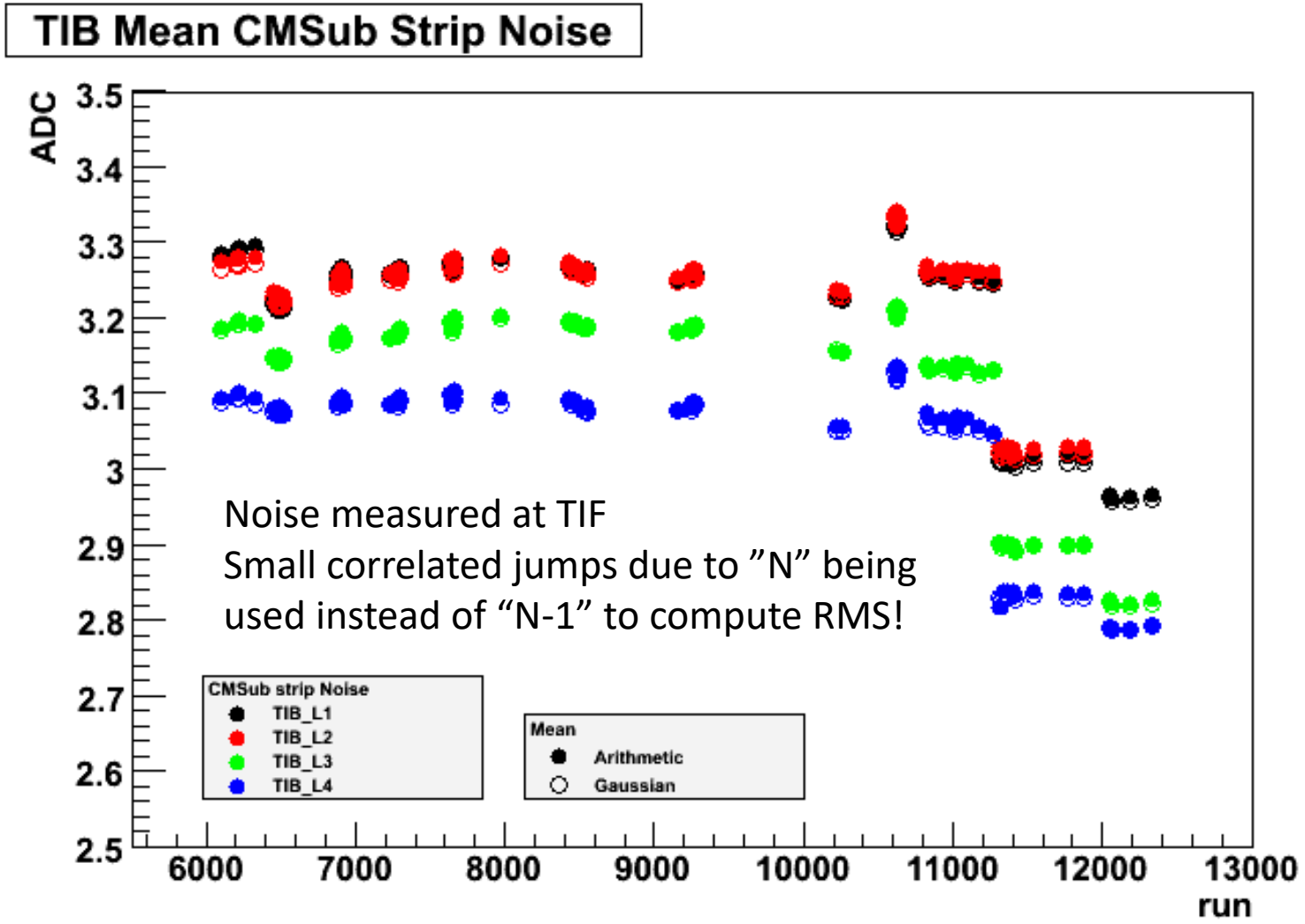
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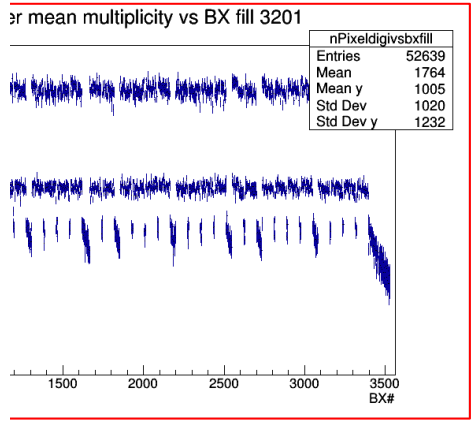
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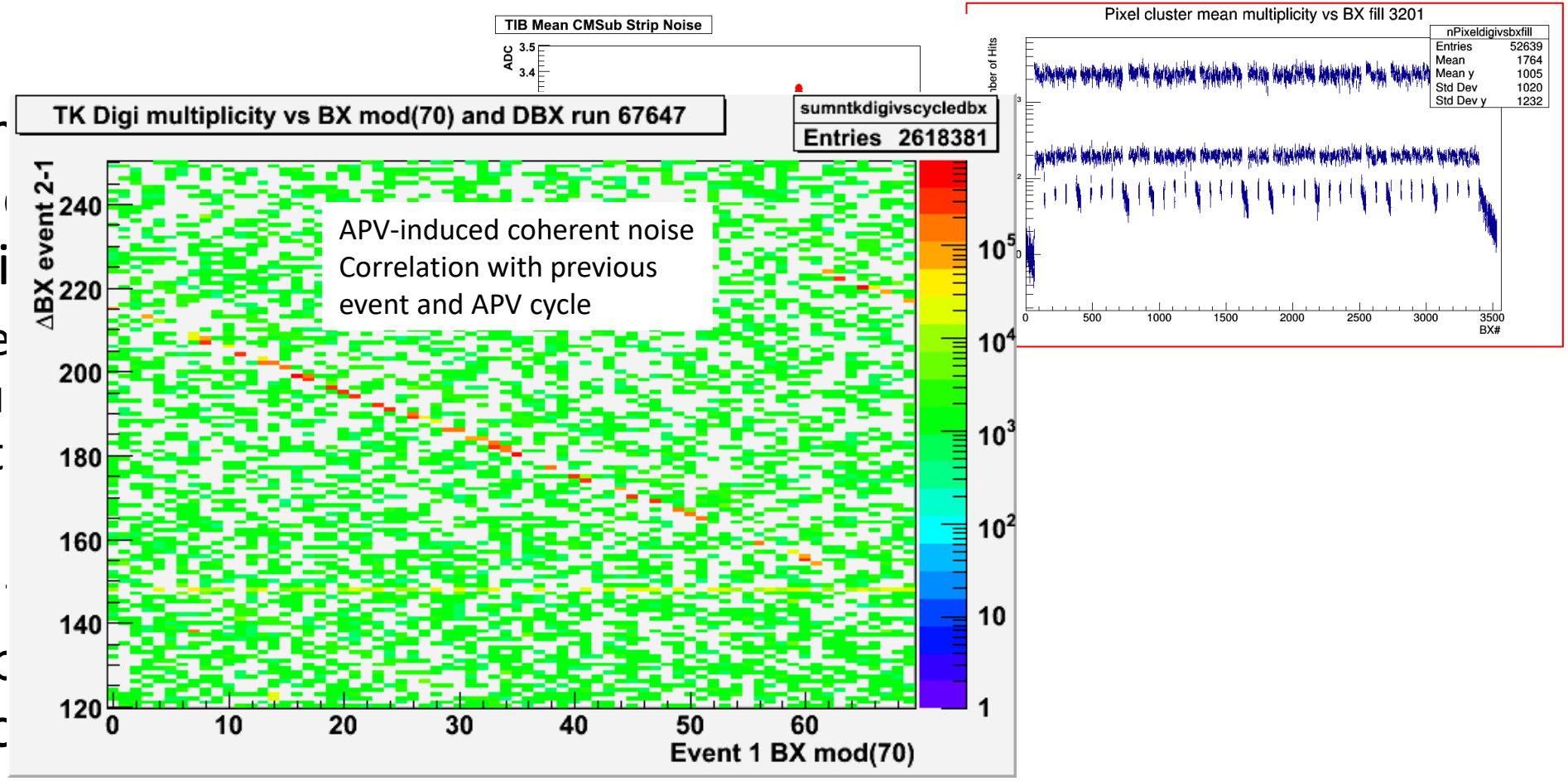
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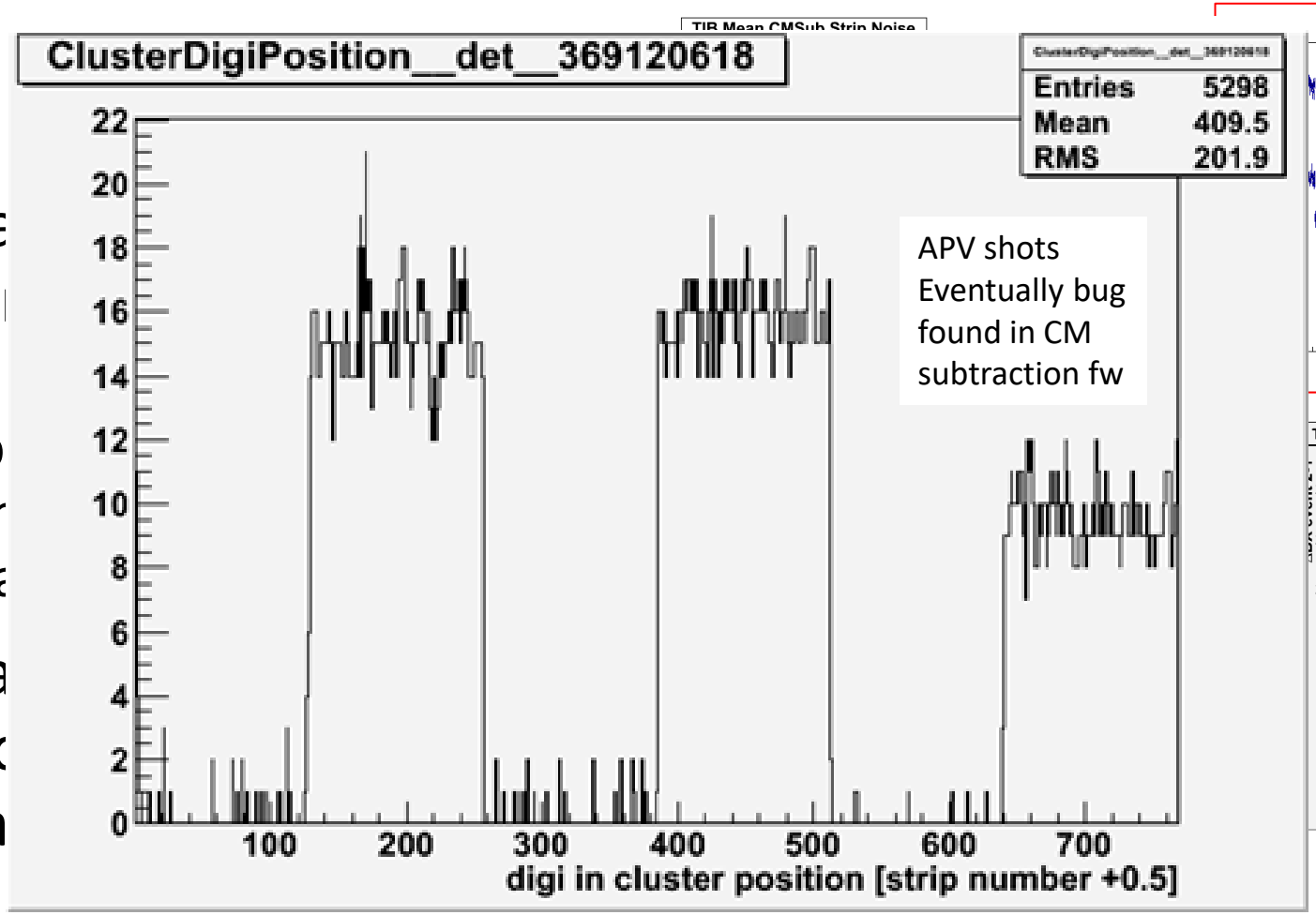
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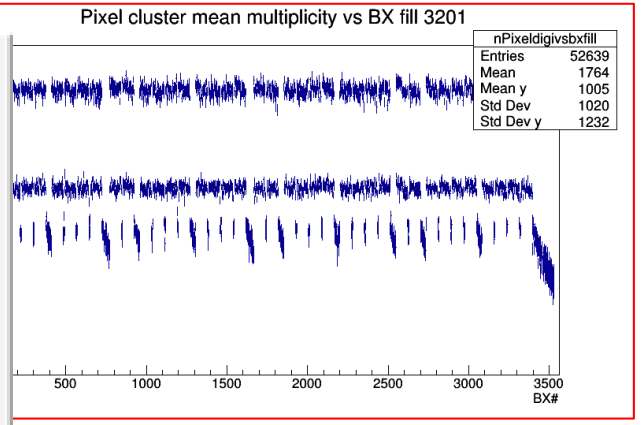
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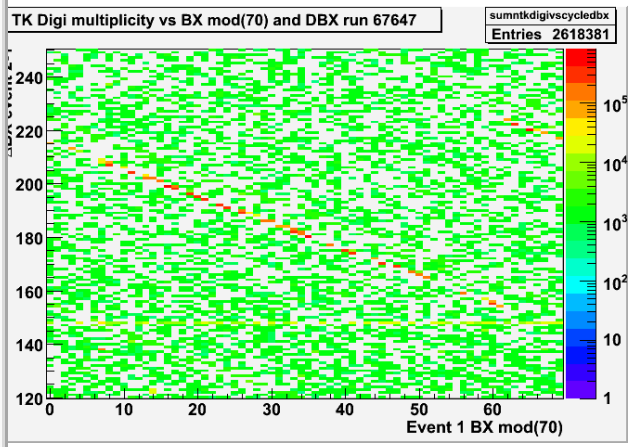


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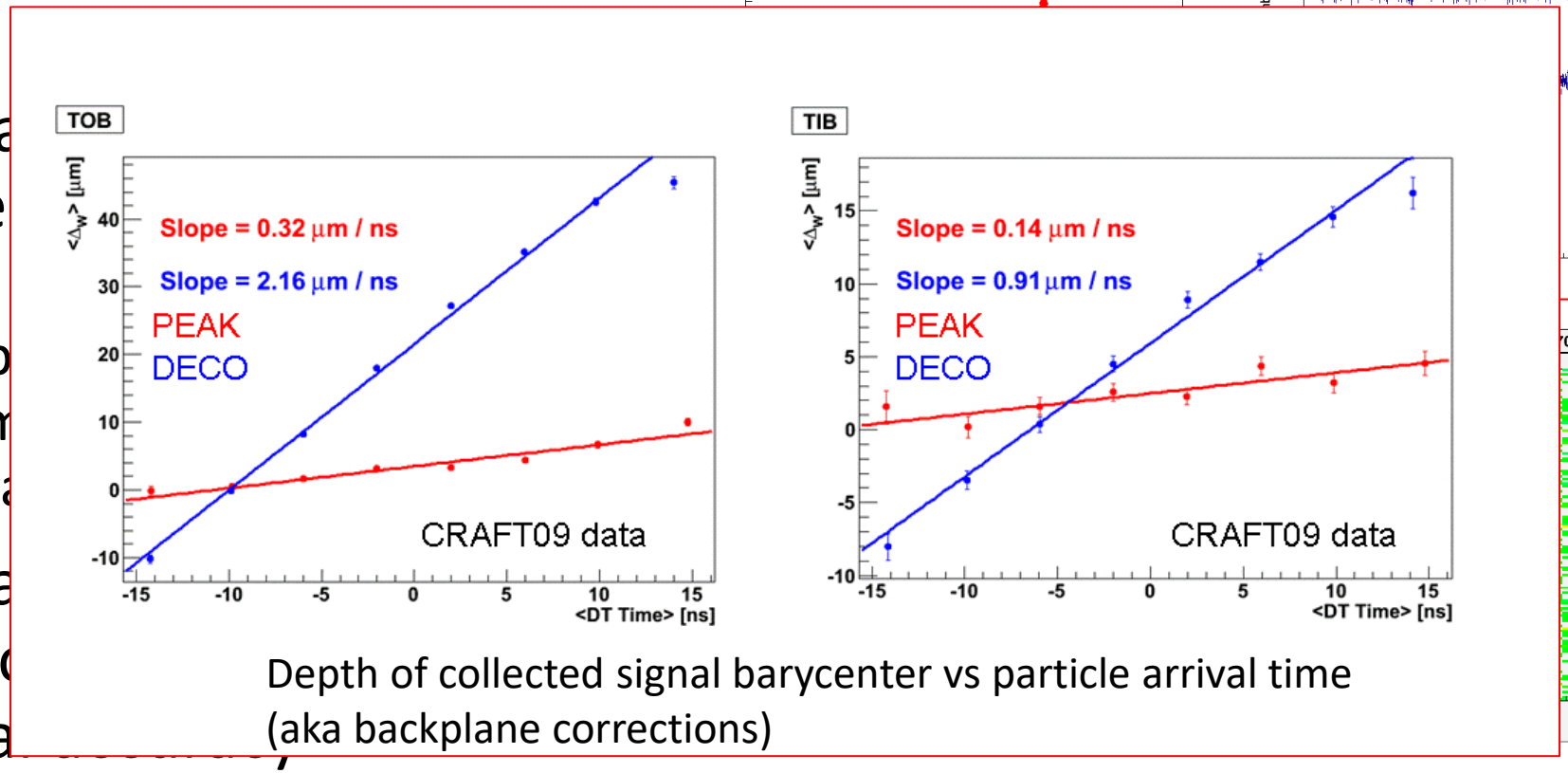




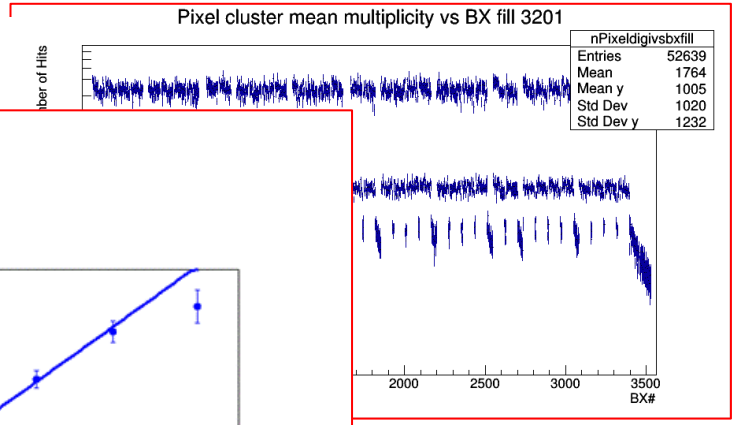


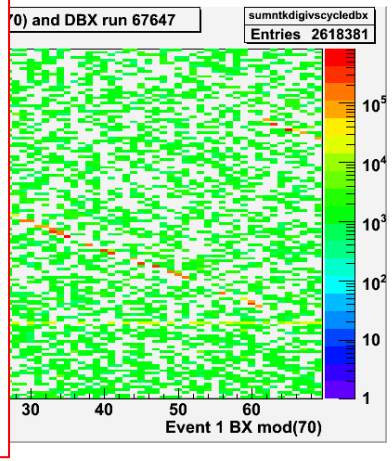
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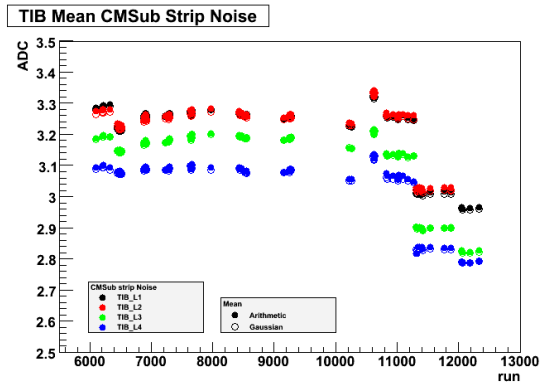
Depth of collected signal barycenter vs particle arrival time
(aka backplane corrections)

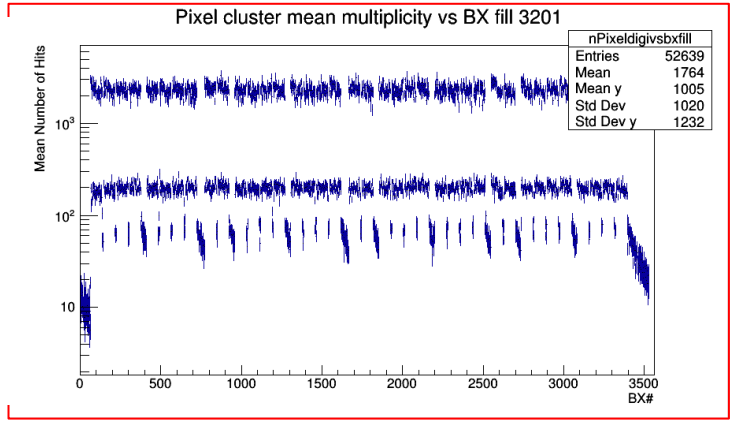


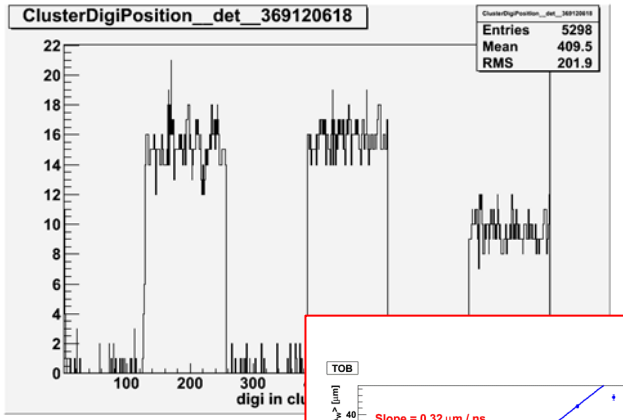


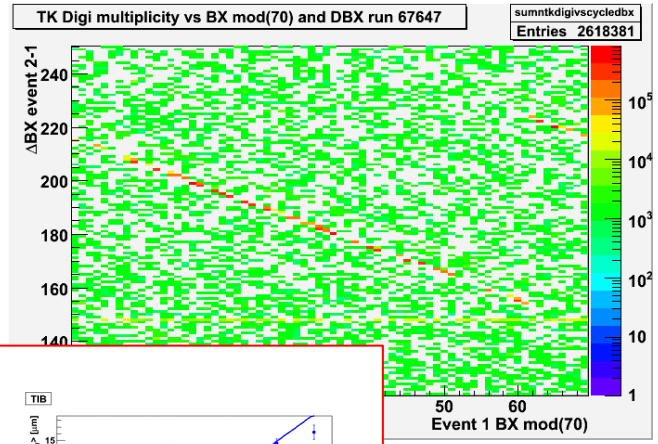
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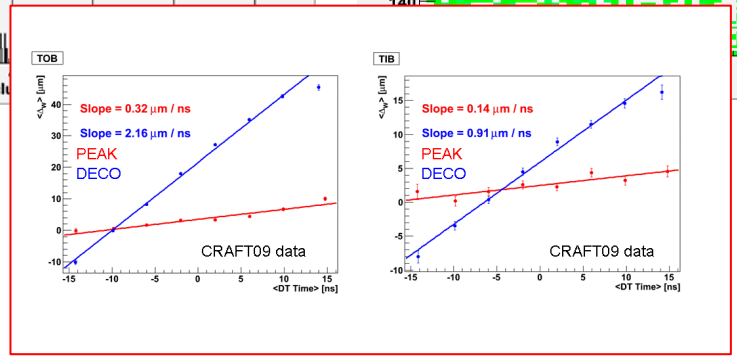
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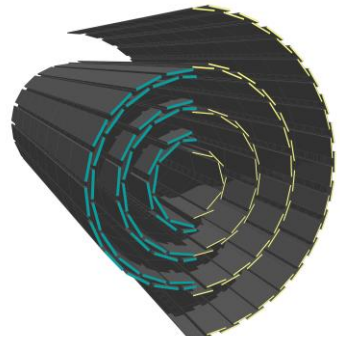




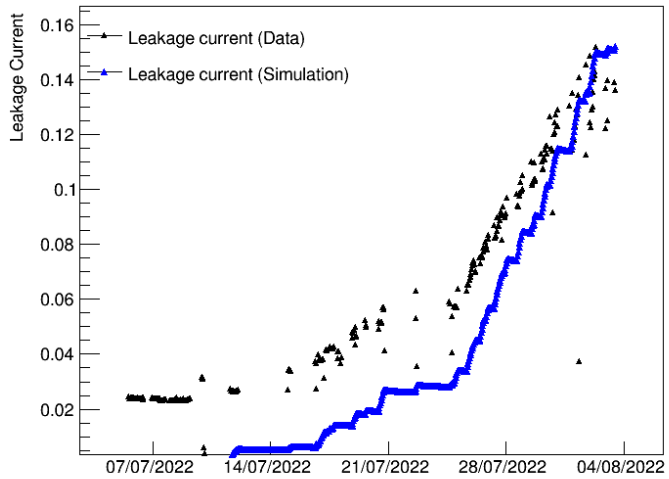


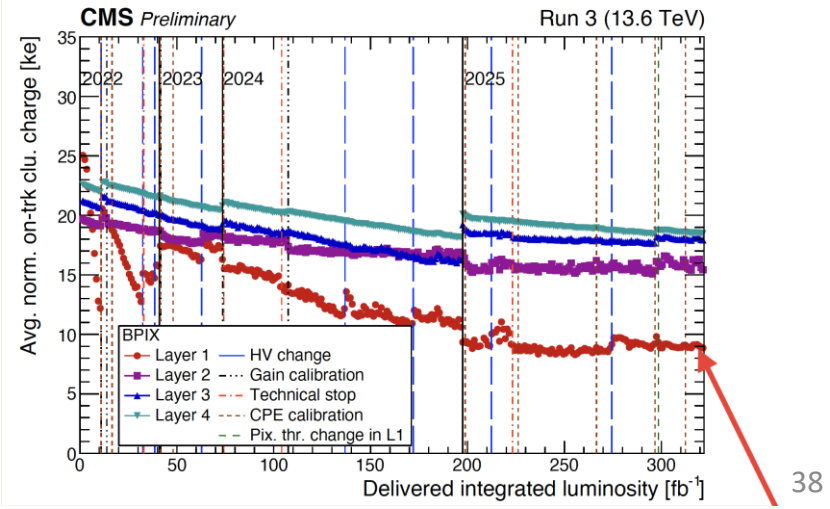
TBPX Layer 1

- CMS Tracker has still an unresolved issue with so close layer 1
  - Legacy pixel detector: R =4.4 cm, Phase-1 and Phase-2: R = 2.7 cm
  - Phase-1 BPIX layer 1 had a troubled story
    - Noisy PROC600, stuck TBMs, non-tunable optimal timing, failing DCDC converters, not understood automasked channels
- It has been and it will be the most affected by SEU and dynamic inefficiencies
- New sensor technology in CMS: 3D sensors
- Not totally clear if radiation estimates are reliable



Leakage Current



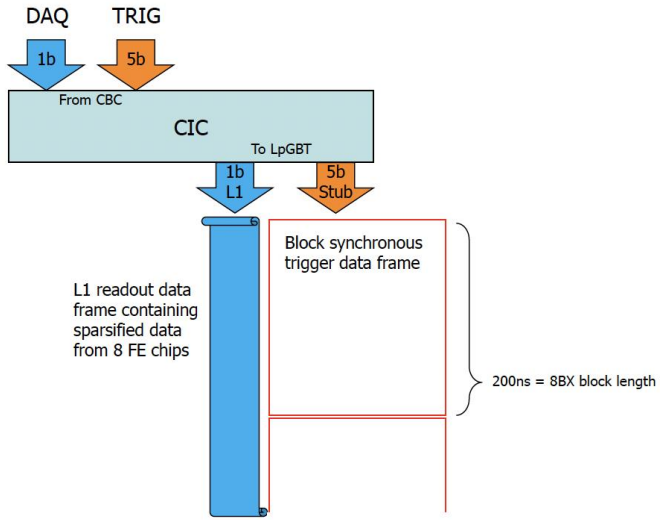


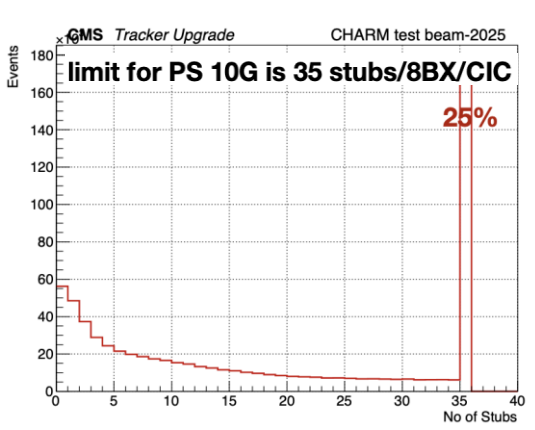
Stubs and L1 Tracks

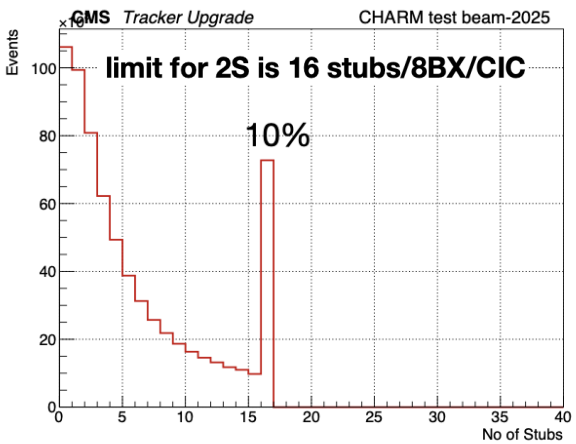
- Contributing to L1 Trigger with continuous stream of data makes understanding of stub efficiency, occupancy, and rate VERY critical
  - Efficiency and noise
    - Stub formation efficiency downstream hit efficiency
  - Stub-specific dynamic inefficiency and data readout limitation
    - Different path wrt triggered event data
  - Where stub data are and when they are available
  - Interplay with L1 Trigger performance
  - Stub offset, search window, and alignment bootstrap
- Excursion: a long-standing wish of mine

Stub readout inefficiency and OOT pileup simulation

- Stub data packed in “boxcars” that take care of 8 consecutive bunch crossings
  - What does not fit is lost
- CMS MC events include the simulation of digis produced by 16 consecutive (pileup) collisions before the event being simulated
- Shall we develop CMSSW code to use those OOT pileup events to emulate and measure stub data losses in a fully consistent simulation?
  - Maybe not to be run in standard production but to produce (in)efficiency maps including correlations among different modules

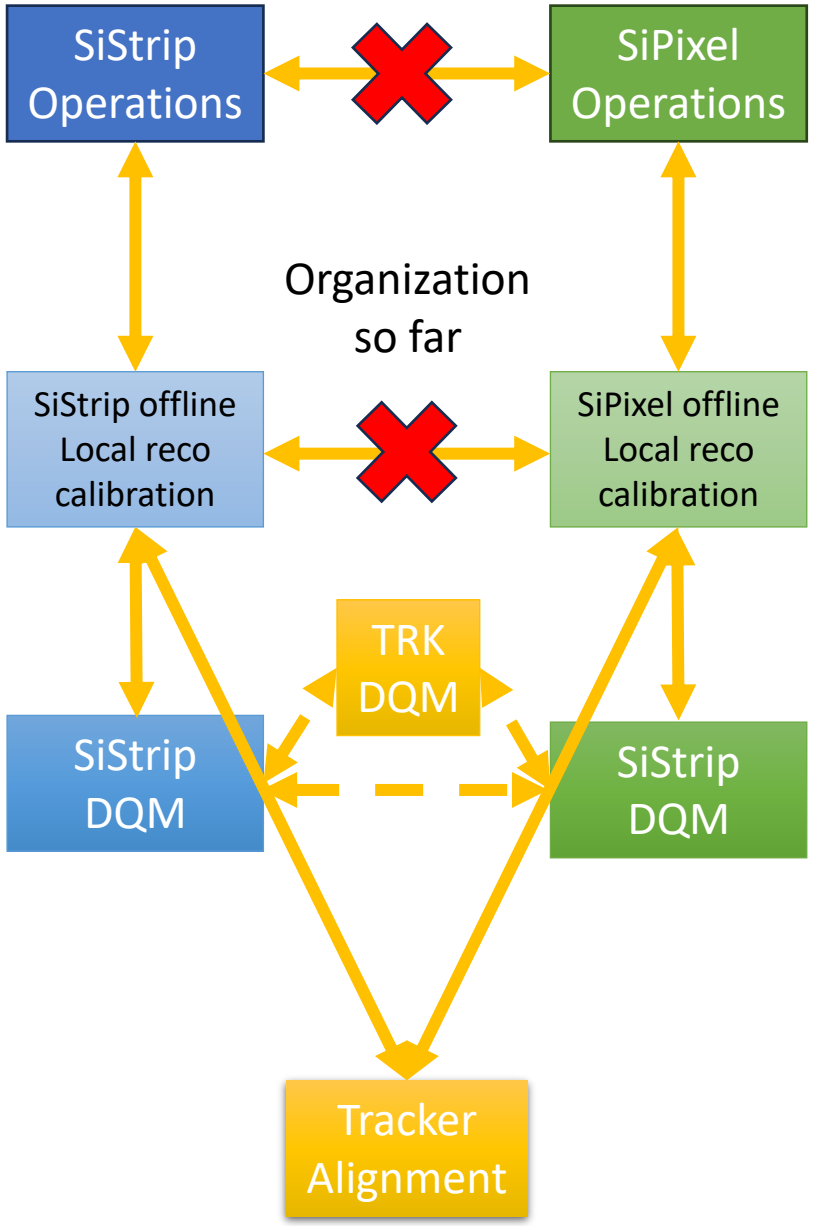






When, Where, By Whom

- Legacy Tracker organize in Operation group(s) and DPG working groups
  - Boundaries frozen eventually
- Tracker Phase-2 organization may be different
  - Tasks and knowledge partly in common between IT and OT already now
  - Improve common software developments among IT and OT
  - Some analyses may (partly) move from offline/online computers to backend boards
  - DQM framework may be different
  - Thresholds on front-end electronics may change analyses priorities and urgency
  - Stub data and L1 Tracks are game changers and bring Tracker in the group of triggering detectors
    - Where and when calibrations for L1 Trigger and HLT
  - Correlation of (offline) data with “slow control”/monitoring data from detector
    - Temperatures, currents, error bits, …
- Discuss and eventually decide how to organize ourselves
- Excursion: basic arithmetic properties



Arithmetic Properties

$$A \cdot B = (A \cdot k)\left(\frac{B}{k}\right)$$

- The pitfall of experimental groups

Arithmetic Properties

$$A \cdot B = (A \cdot k)\left(\frac{B}{k}\right)$$

$$A + B = (A + \delta) + (B - \delta)$$

- The pitfall of experimental groups

Arithmetic Properties

$$A \cdot B = (A \cdot k)\left(\frac{B}{k}\right)$$

Calibration factors,
Scale factors,
…

$$A + B = (A + \delta) + (B - \delta)$$

Bias corrections,
Alignment parameters,
…

- The pitfall of experimental groups

Arithmetic Properties

$$A \cdot B = (A \cdot k)\left(\frac{B}{k}\right)$$

Calibration factors,
Scale factors,
…

$$A + B = (A + \delta) + (B - \delta)$$

Bias corrections,
Alignment parameters,
…

- The pitfall of experimental groups: do not argue on k or δ !

Arithmetic Properties: Corollary #1

$$A \cdot B = (A') \left(\frac{A \cdot B}{A'} \right)$$

$$A + B = (A') + (A + B - A')$$

- The pitfall of experimental groups: do not argue on k or δ !
  - Sometimes what comes later undo what done earlier

Arithmetic Properties: Corollary #1

$$A \cdot B = (A') \left(\frac{A \cdot B}{A'} \right)$$



$$A + B = (A') + (A + B - A')$$

- The pitfall of experimental groups: do not argue on k or δ !
  - Sometimes what comes later undo what done earlier

Arithmetic Properties: Corollary #2

$$A \cdot B \neq (A \cdot k)\left(\frac{B}{k'}\right)$$

$$A + B \neq (A + \delta) + (B - \delta')$$

- The pitfall of experimental groups: do not argue on k or δ !
  - This is the REAL problem!
  - Often the problem is not the definition of the plan but a consistent execution of it
    - Time dependent template or Lorentz Angle corrections vs time dependent alignment deployments, …

Arithmetic Properties

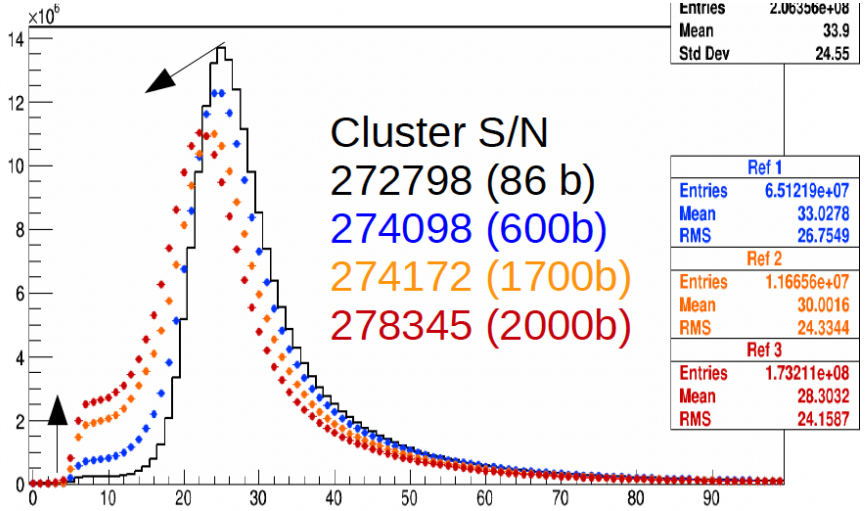
$$A \cdot B = (A \cdot k)\left(\frac{B}{k}\right)$$

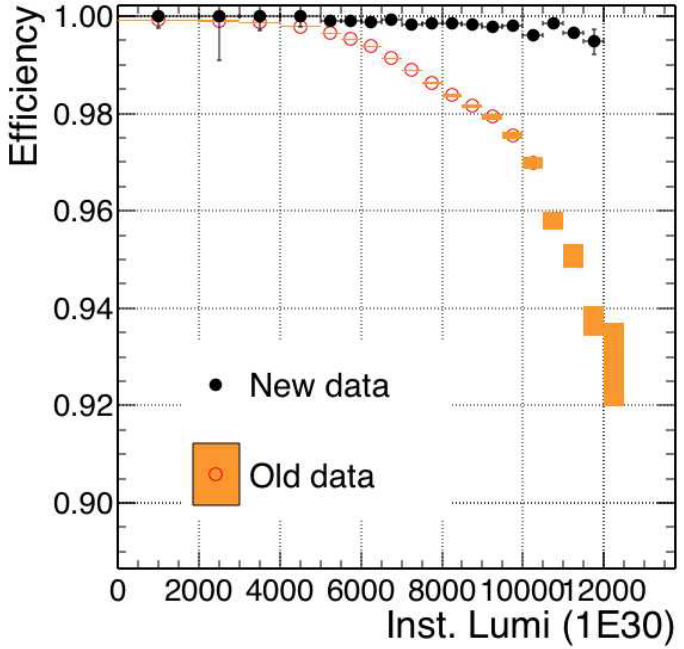
$$A + B = (A + \delta) + (B - \delta)$$

- Definition of k or δ does not matter, monitoring them does
  - Linked to group(s) and tasks organization
  - Excursion: my biggest/only regret

My Biggest/Only Regret

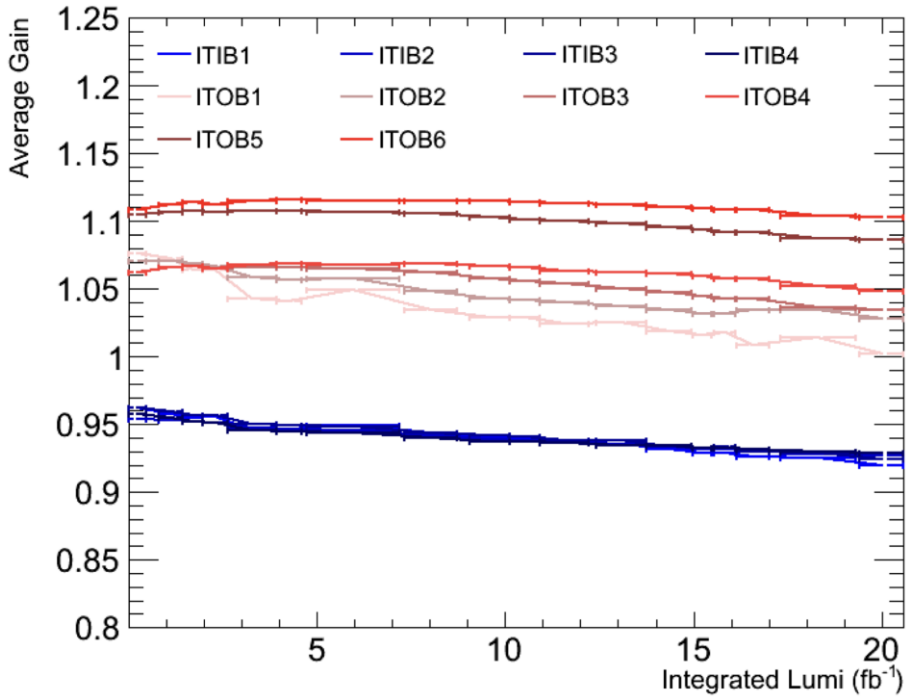
- 2016 has been a troubled year for SiStrip Tracker
  - Higher luminosity and pileup caused saturation of APV and loss of signal
- A fraction of 2016 data are normally not used in physics analysis because of poor SiStrip Tracker efficiency
- Problem solved in Summer 2016 by Erik Butz who suggested a different setting of VFP parameter of the APVs
  - No more saturation no more signal and efficiency loss





My Biggest/Only Regret

- Hints of the problem were visible already in 2012 data
  - Luminosity has been increased sizably
- Observed in “G2” calibration workflow data
  - Misinterpreted as an integrated luminosity related signal loss
- Never understood the cause/effect link
  - And yet not pursued alternative analysis which could have revealed the dependence on instantaneous luminosity
- Problem could have been solved in 2015 with no data loss in 2016
- I was DPG coordinator and I regret it
- Do not stop when you have completed a measurement: understand it! Past and future Trackers are complicated



Alignment

- New Geometry/layout
  - To be understood if there are critical deformations
  - Ensure that tracks catch as many as possible hits in the overlaps
  - Robust pixel-only forward tracking is a must otherwise TFPX and TEPX will not be exploited
- New mechanical structures
  - Verify and monitor if they move with time and/or correlated with external conditions (mostly temperature and B field)
  - Default assumption: alignment does not need systematic surveys during assembly to bootstrap. Is it ok?

Preparation for first beam

- Much less time than before Run 1
  - Steeper learning curve required
- No year-long CRUZET and/or CRAFT
  - Different cosmic/collision ratio for alignment
- To be done
  - Exercise commissioning, monitoring and alignment tools in advance with simulated data
  - Estimate time and data needed for the alignment
- Speak up if time is not enough!
  - CMS may have forgotten how much time was needed to prepare Run 1

Conclusions

- Focus on
  - Monitoring performance of Outer Tracker Stubs
  - Robust hit efficiency measurements
  - Characterization of new sensors and new readout chips
  - Binary readout threshold tuning
  - New layout and mechanics and readiness of alignment
  - Robust tracking in Inner Tracker forward extension
  - Preparing tools for compressed commissioning
  - Radiation effect simulations
  - Exploiting MTD
  - Sensible sharing of tasks and tools with operation teams